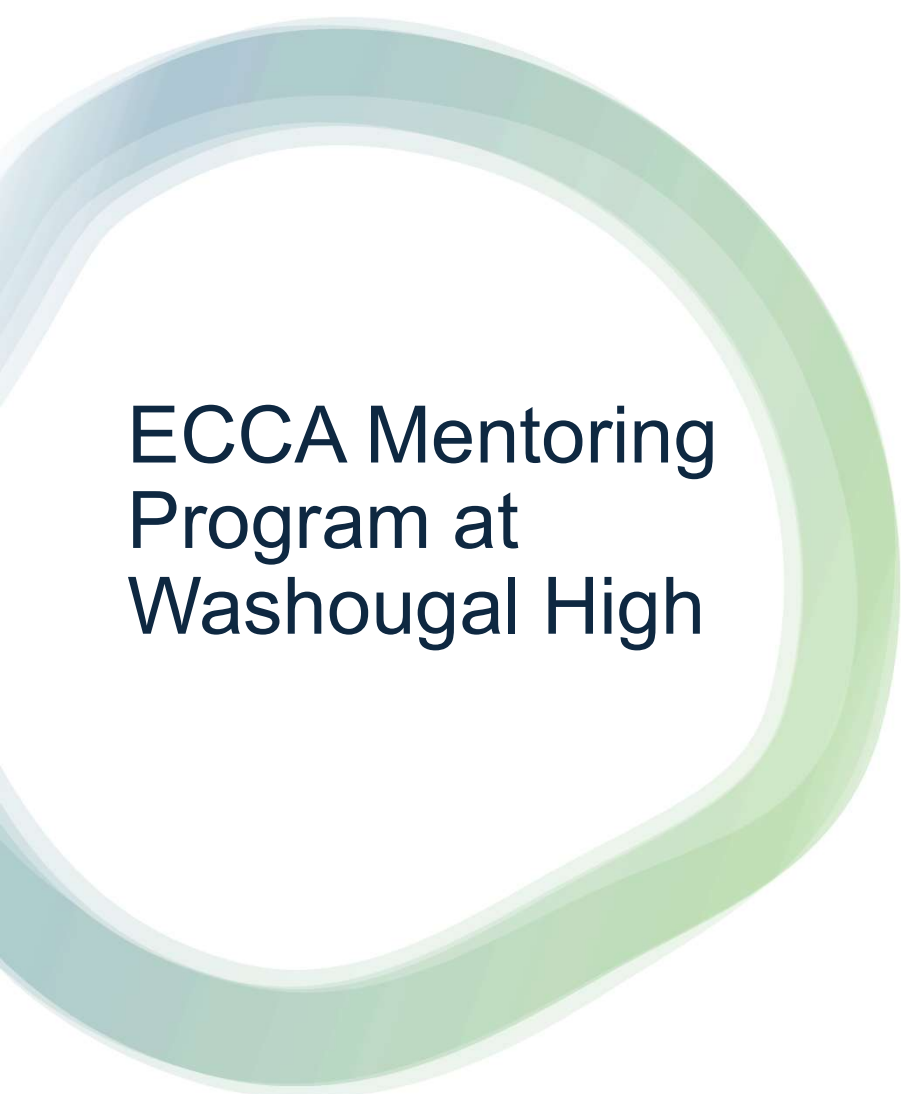




ECCA Mentoring Program at Washougal High

- **Team Data Wizards:**
- Kaj Hansteen Izora
- Tommy Hoang-Tien
- Kira Novitchkova-Burbank
- Kimberly Rodriguez Chihuahua



Outline

Introduction

Data Preprocessing

Data Exploration

Methodology

Results

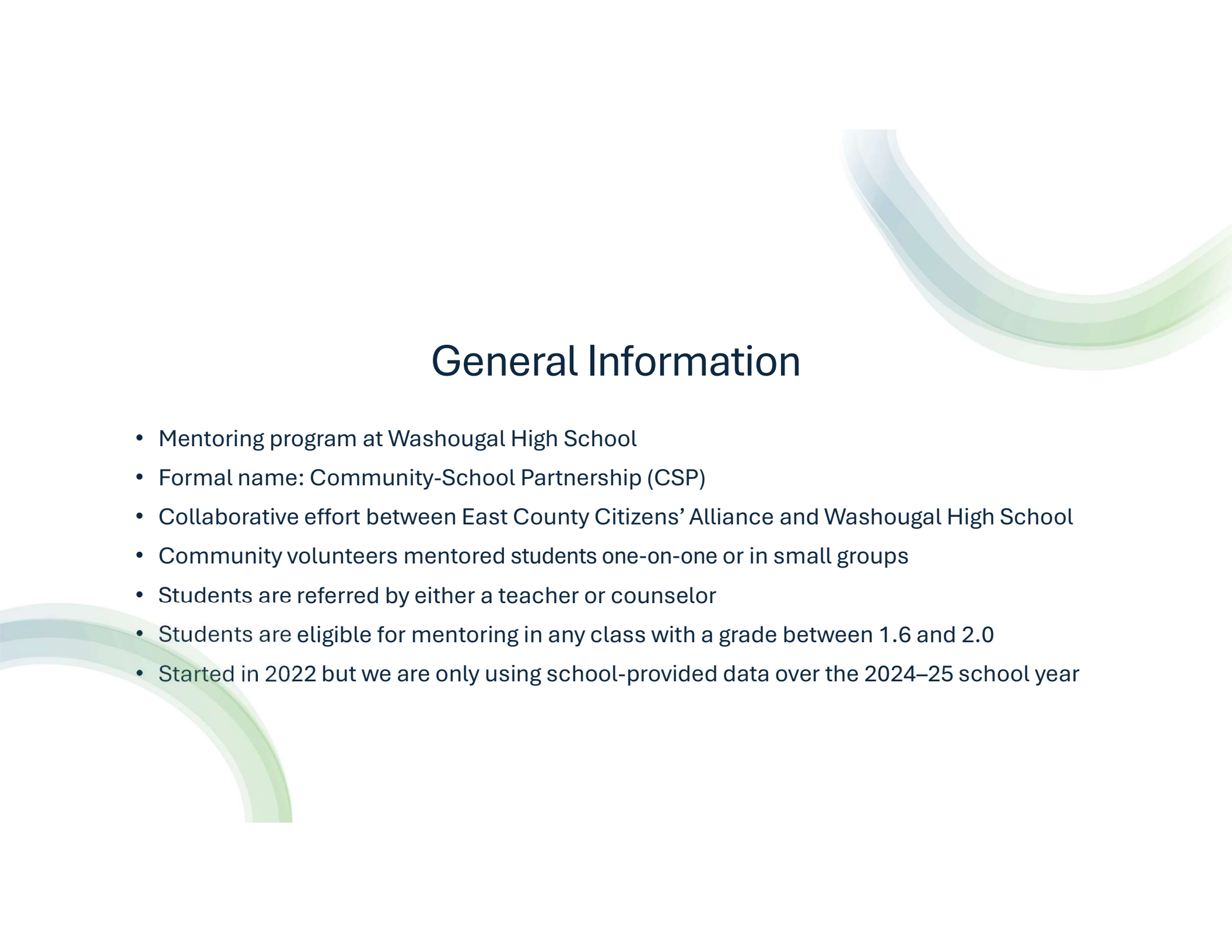
Conclusion

Questions

Appendix

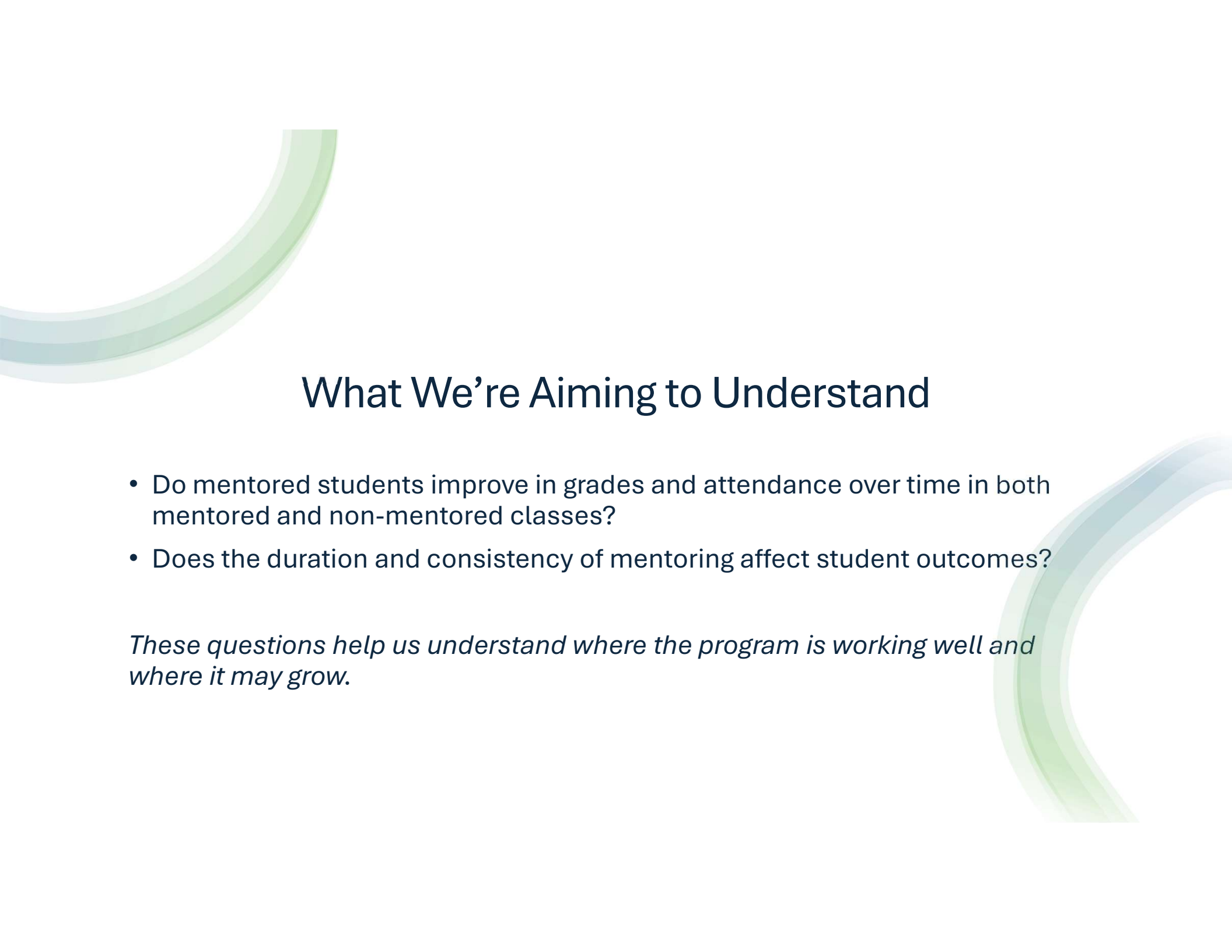


Introduction



General Information

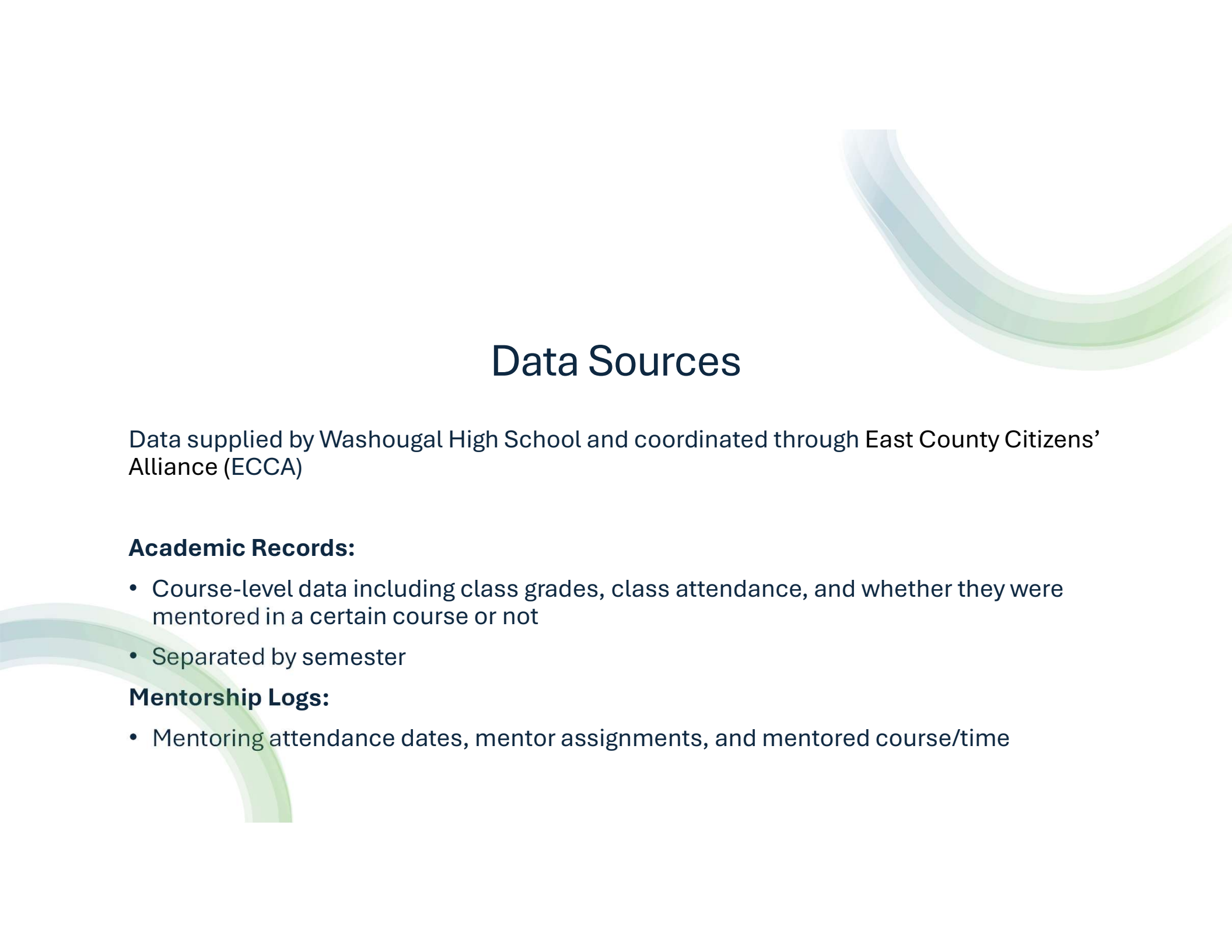
- Mentoring program at Washougal High School
- Formal name: Community-School Partnership (CSP)
- Collaborative effort between East County Citizens' Alliance and Washougal High School
- Community volunteers mentored students one-on-one or in small groups
- Students are referred by either a teacher or counselor
- Students are eligible for mentoring in any class with a grade between 1.6 and 2.0
- Started in 2022 but we are only using school-provided data over the 2024–25 school year



What We're Aiming to Understand

- Do mentored students improve in grades and attendance over time in both mentored and non-mentored classes?
- Does the duration and consistency of mentoring affect student outcomes?

These questions help us understand where the program is working well and where it may grow.



Data Sources

Data supplied by Washougal High School and coordinated through East County Citizens' Alliance (ECCA)

Academic Records:

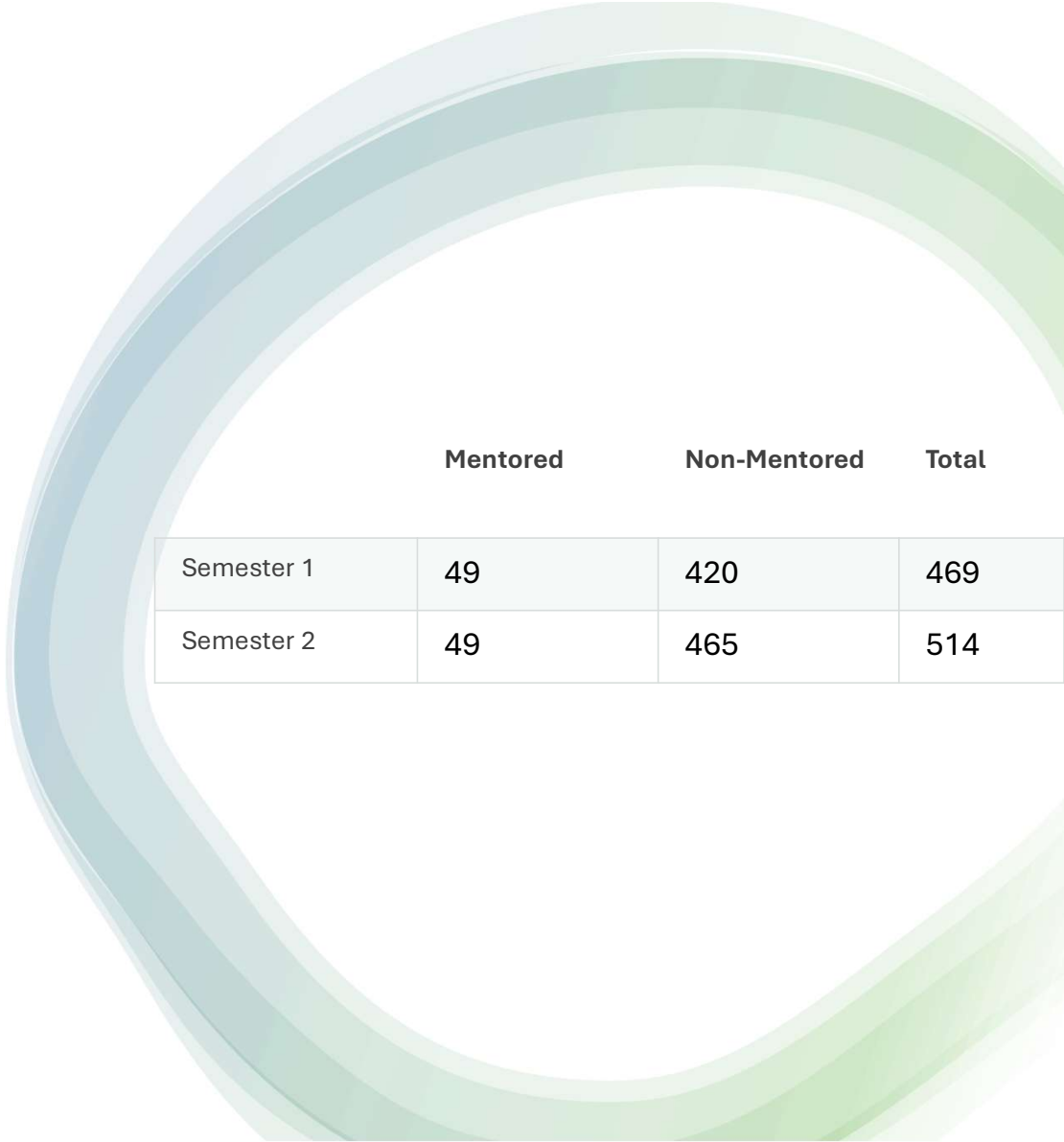
- Course-level data including class grades, class attendance, and whether they were mentored in a certain course or not
- Separated by semester

Mentorship Logs:

- Mentoring attendance dates, mentor assignments, and mentored course/time

More Specifics on the Data Given

- 29,830 rows in cleaned dataset
- 28 columns in cleaned dataset
- 14 mentors across both semesters
- 118 unique courses over both semesters



	Mentored	Non-Mentored	Total
Semester 1	49	420	469
Semester 2	49	465	514

The background of the slide features a light green gradient. On the left side, there are several overlapping, wavy, semi-circular shapes in shades of blue and teal. On the right side, there are similar wavy shapes in shades of light green and yellow-green.

Data Preprocessing

Overview of Data Cleaning Goals

Overview of Data Cleaning Goals

Why Cleaning Was Necessary:

- Data came from multiple sources (academic records & mentoring logs)
- Needed to ensure clean structure before analysis

Key Tasks We Tackled:

- Joined all datasets into one final dataset
- Filled in any values not provided
- Preserved missing data where appropriate (e.g., no grade yet)
- Created new columns for analysis

Key Data Cleaning Steps

Merged Grades with Mentoring Logs

- Aligned weekly grades with mentoring sessions
- Added flags: attended that week, ever attended, mentored course

Filled in Not Provided Weeks

- Ensured every student-course pair had Weeks 2–20 (semester 1) and 21–34 (semester 2)
- Added rows with “No” for mentoring and “No Grade” if not provided

Other Fixes

- Removed any duplicates, preserved missing values
- Standardized formats and double checked for no manual entry issues

Key Variables Added During Cleaning

Z-Score (Standardized Grade/Absents)

- Allows us to compare student performance relative to their peers week by week
- This is used as a response variable in our model

Running Total of Mentoring Sessions/Time Since Last Mentored

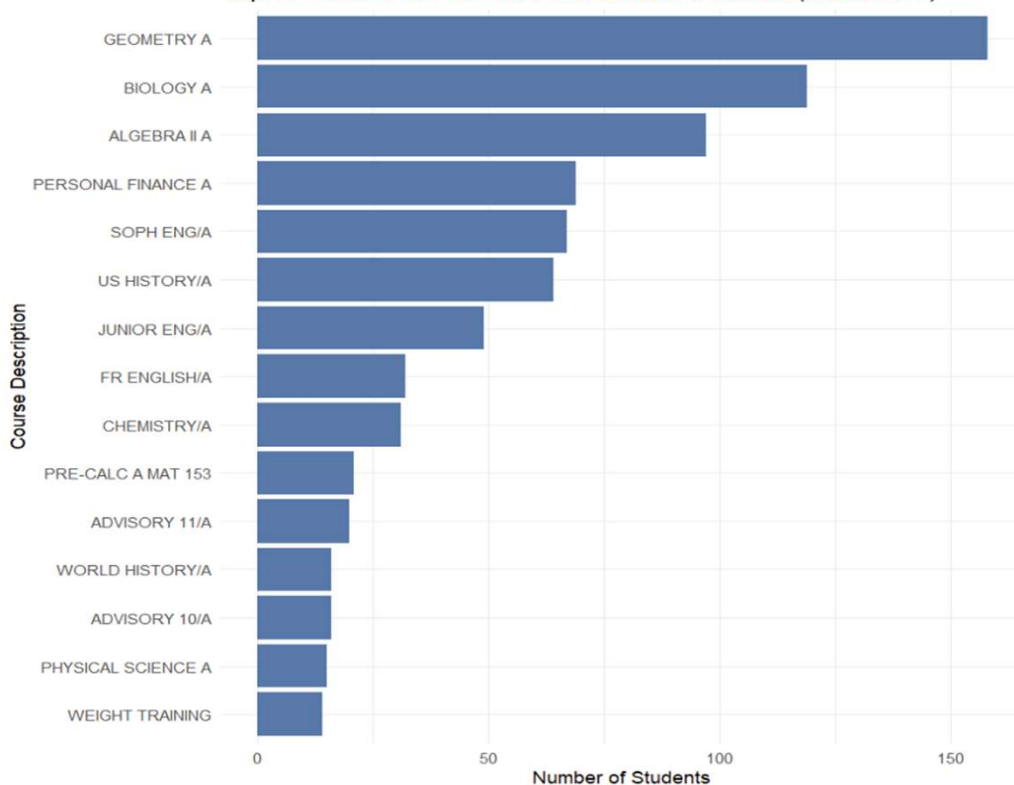
- Helps assess whether more/recent/sustained mentoring correlates with better academic outcomes
- These variables are used as predictors in our model to examine how mentoring influences student performance over time.



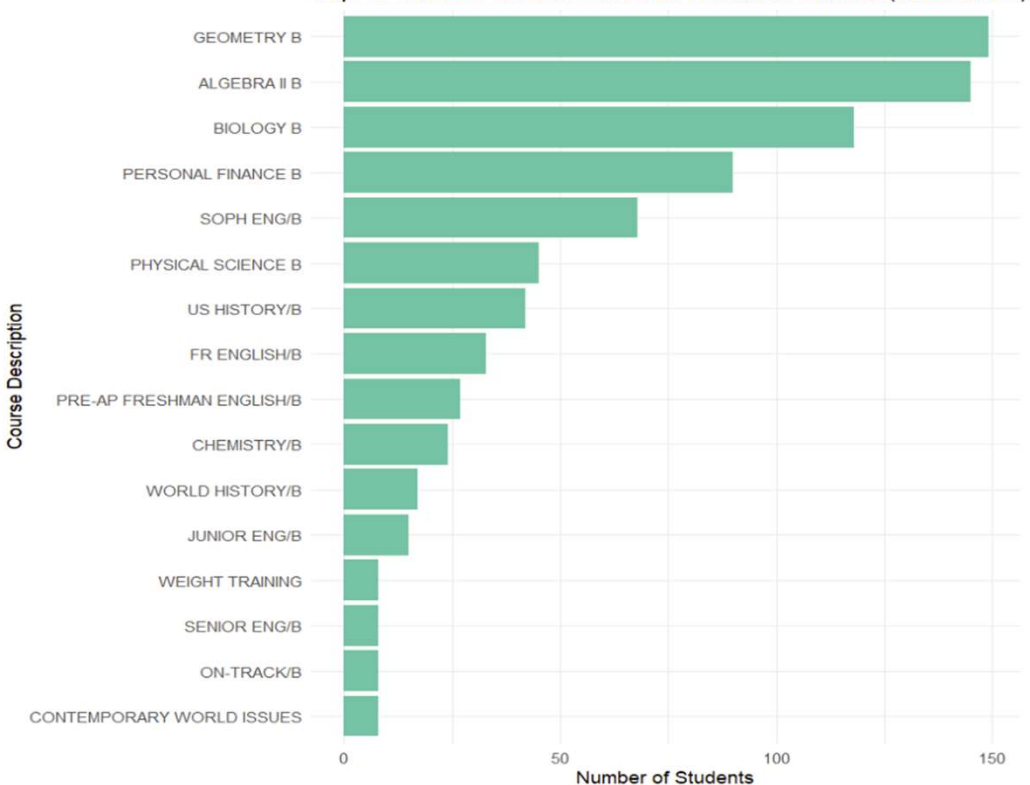
Data Exploration

What are the most commonly taken courses at the school with at least 1 mentored student?

Top 15 Courses with At Least One Mentored Student (Semester 1)

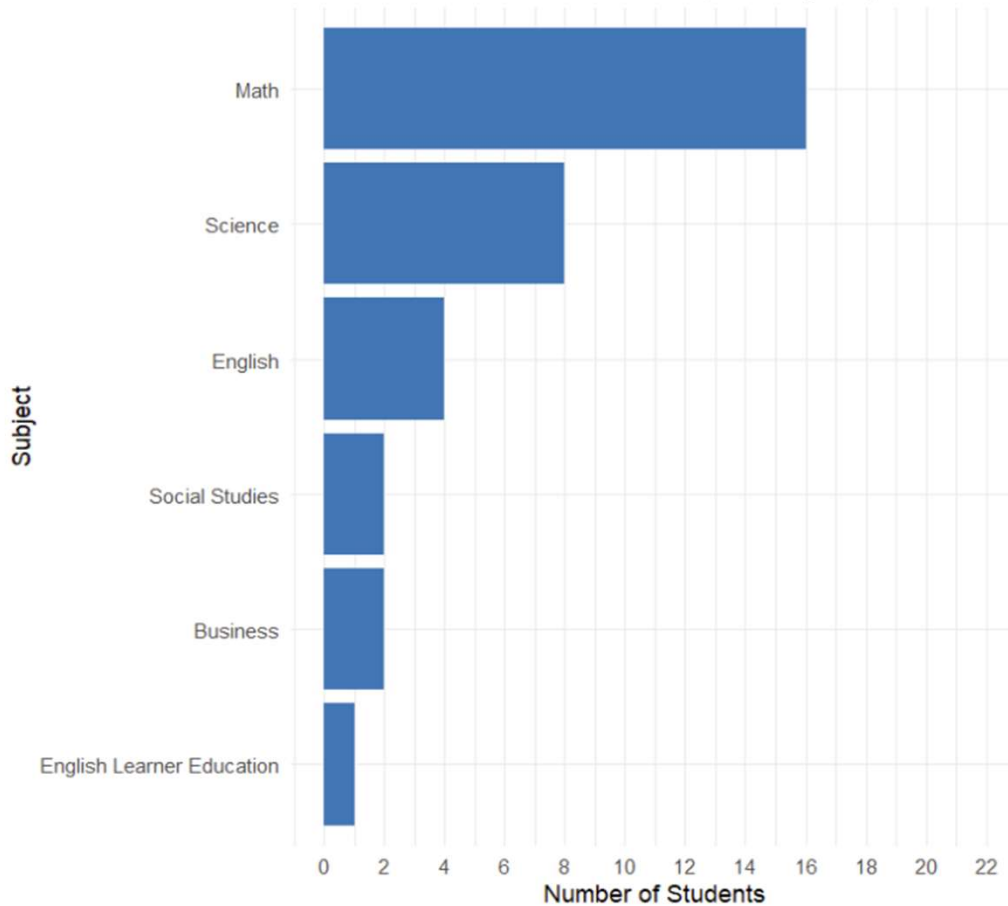


Top 15 Courses with At Least One Mentored Student (Semester 2)

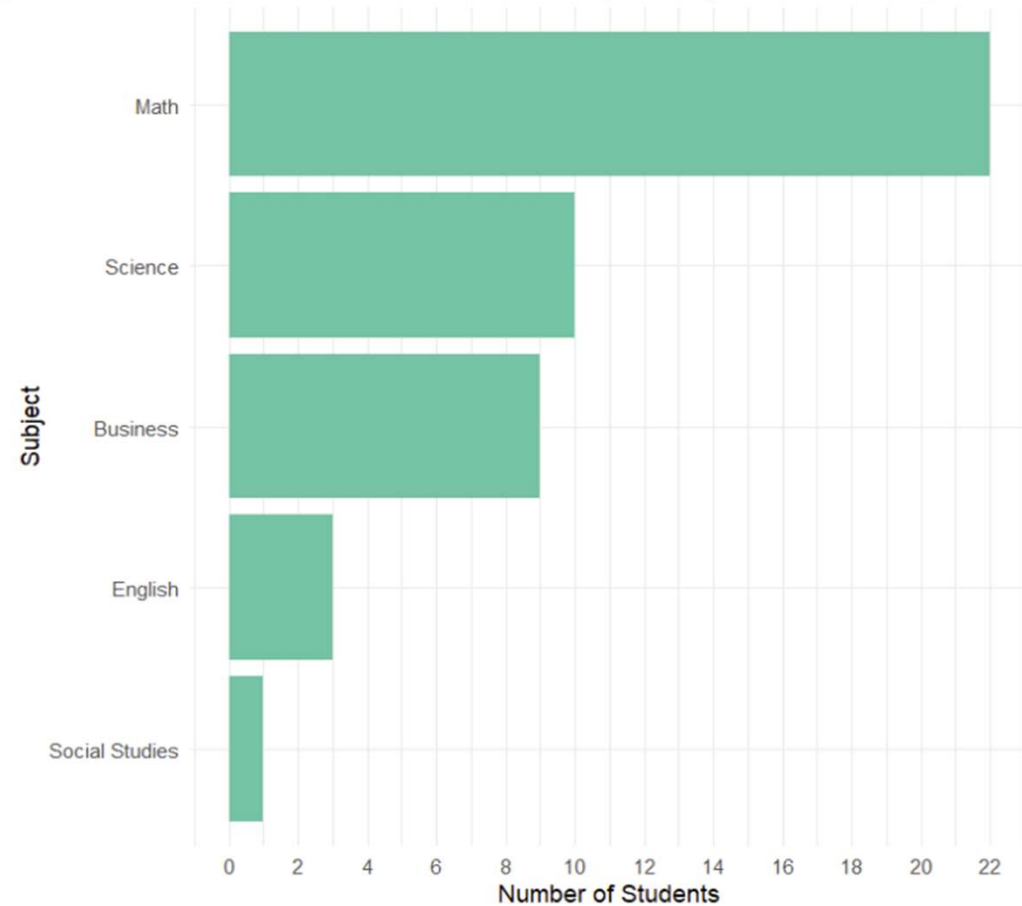


How many students get mentored in each subject?

Number of Mentored Students per Subject (Semester 1)

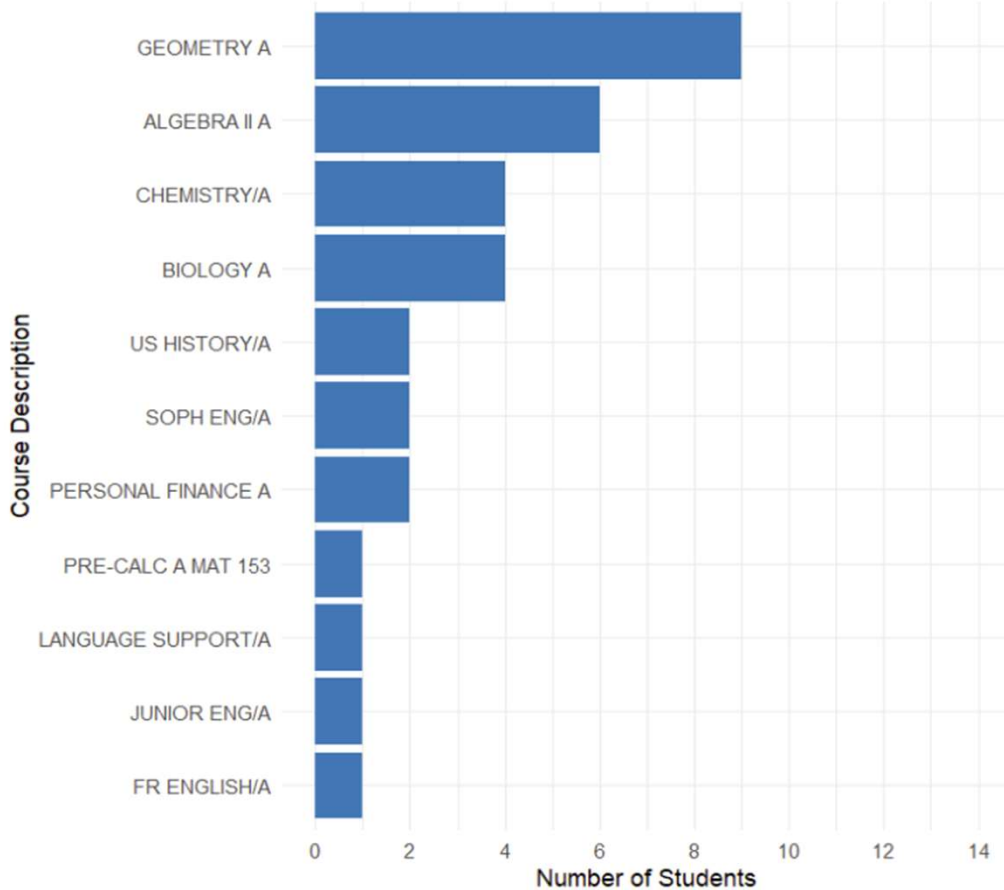


Number of Mentored Students per Subject (Semester 2)

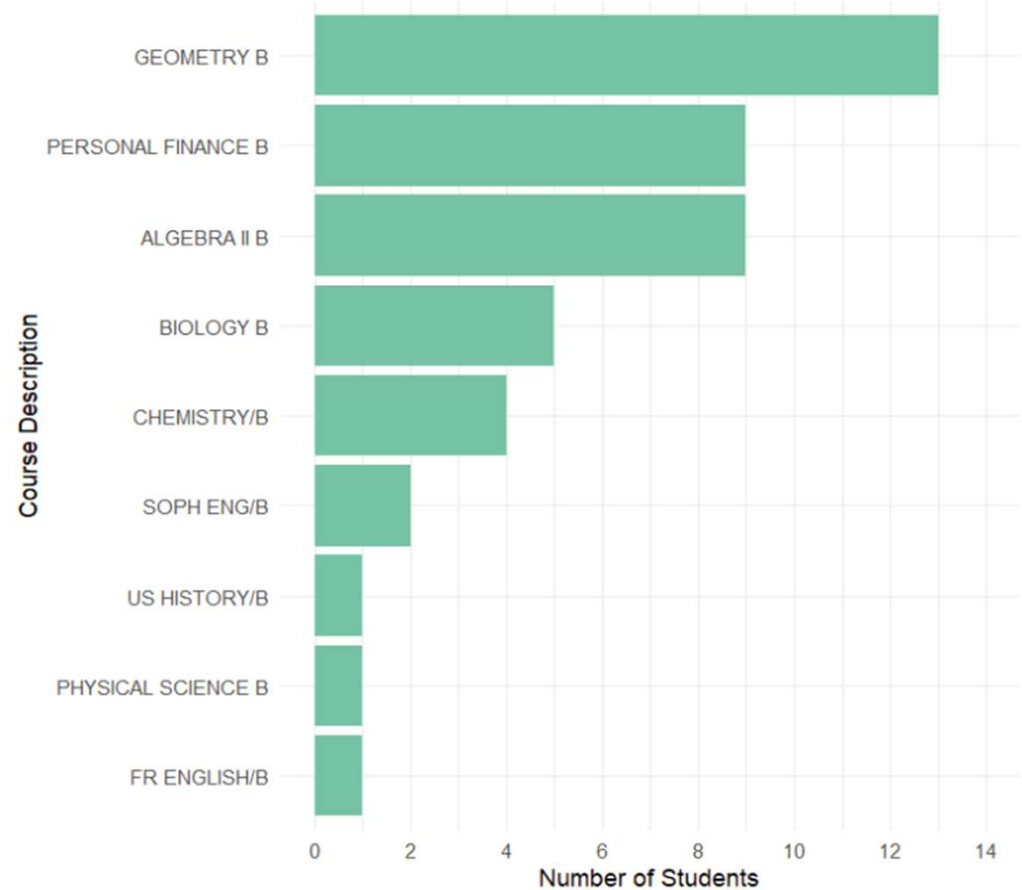


How many students get mentored in each course?

Number of Mentored Students per Course (Semester 1)



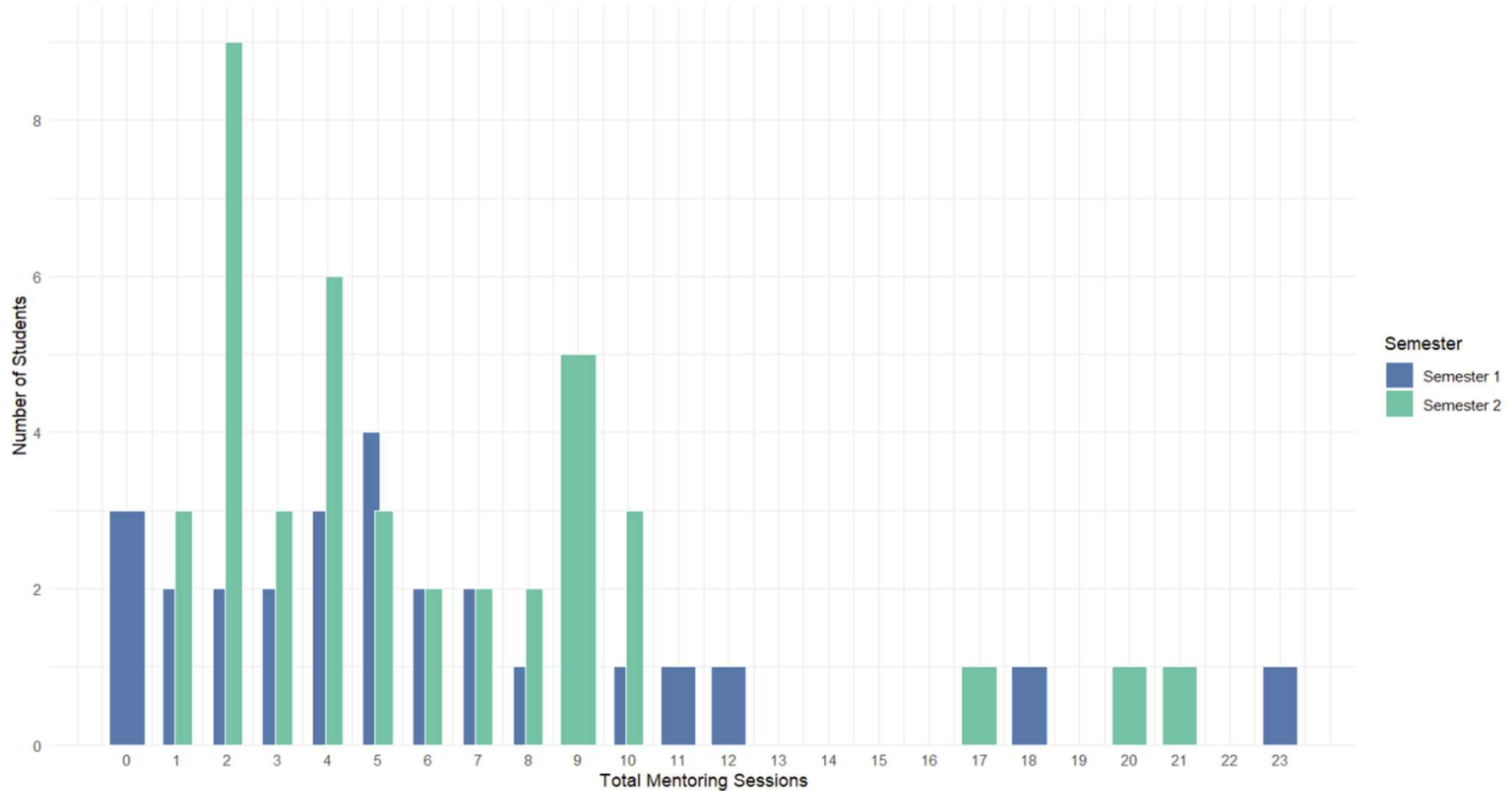
Number of Mentored Students per Course (Semester 2)



How many mentoring sessions do students typically attend per semester?

Total Mentoring Sessions Attended by Mentored Students per Semester

Grouped bars show number of students per session count, by semester

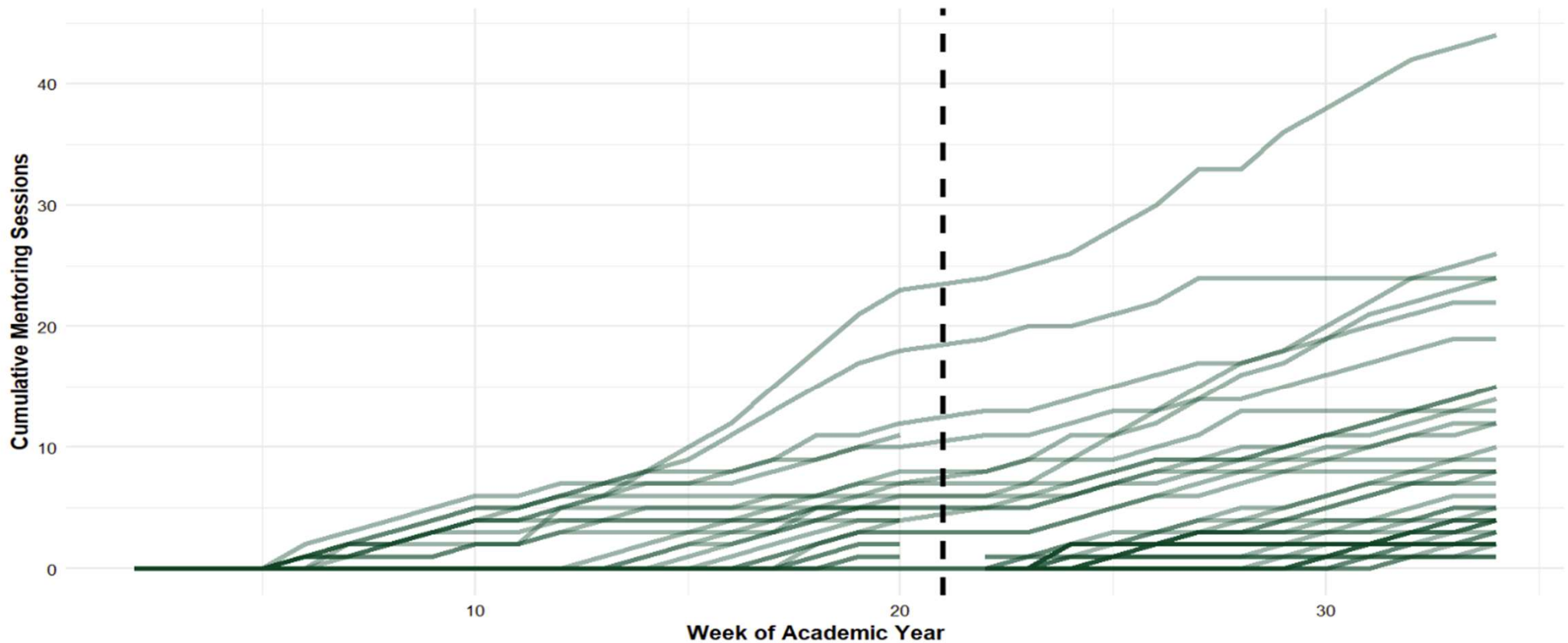


How often did mentored students attend sessions over the academic year?

Cumulative Mentoring Sessions Attended by Mentored Students

Each line represents one mentored student's cumulative sessions

■ Cutoff Between Semesters 1 & 2

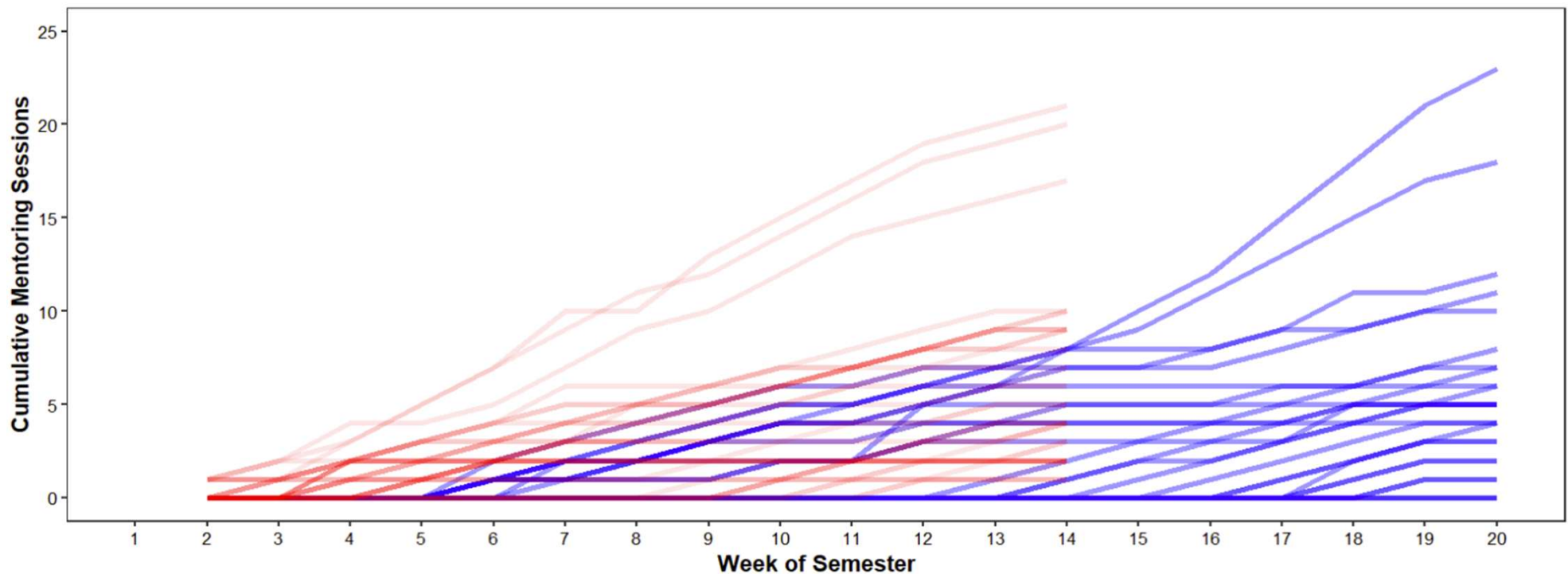


How does mentoring attendance differ between Semester 1 and Semester 2?

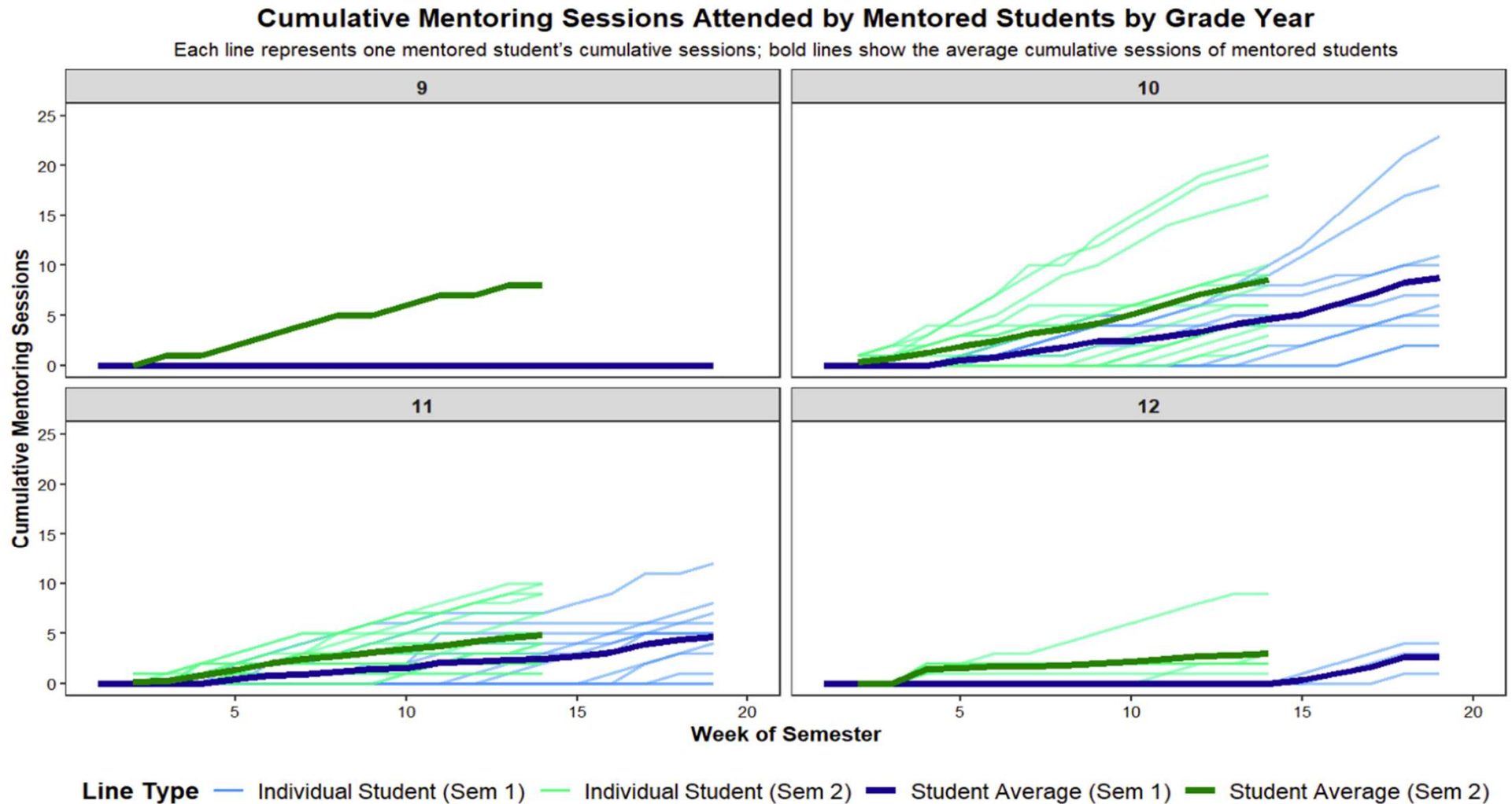
Cumulative Mentoring Sessions Attended by Mentored Students

Each line represents one mentored student's cumulative sessions

Semester — Semester 1 — Semester 2



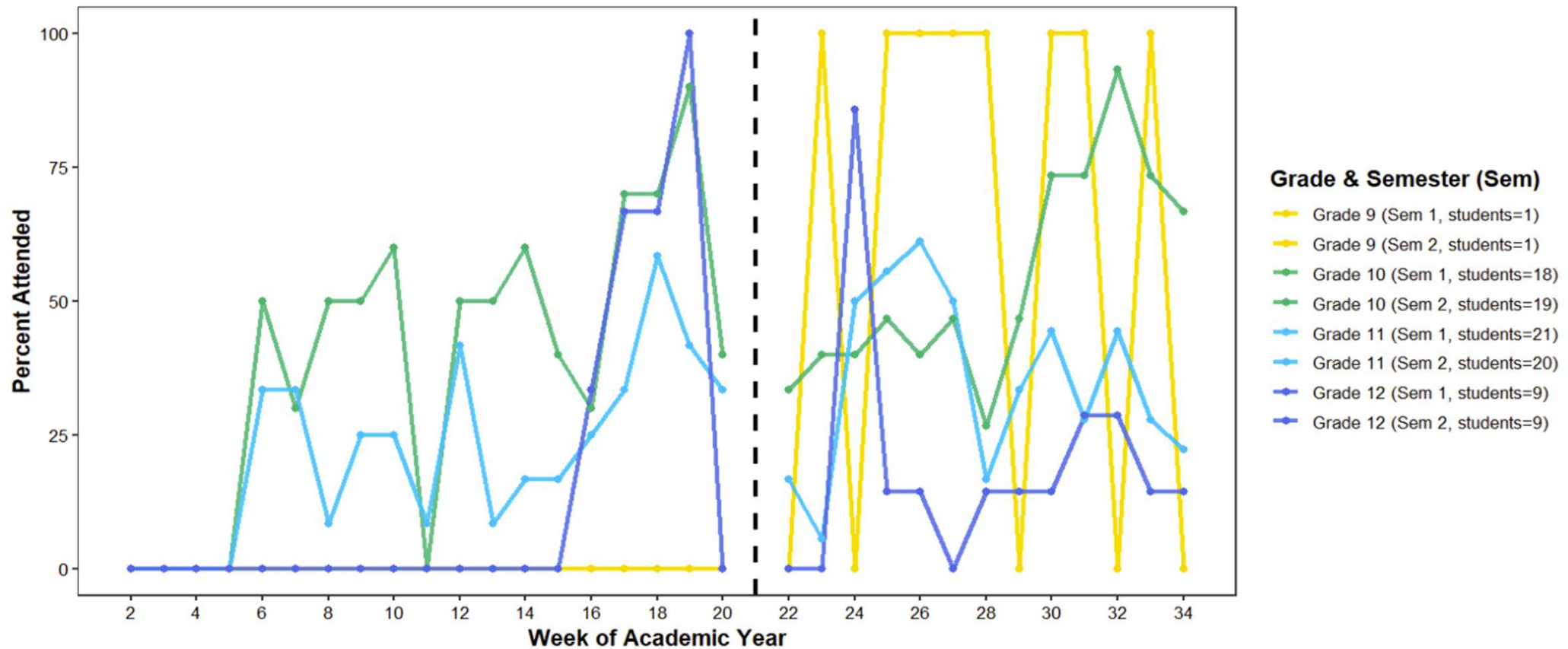
How does mentoring attendance vary by grade year?



What percent of mentored students attend mentoring each week by grade year?

Weekly Percentage of Mentored Students Attending at least 1 Mentoring Session by Grade

Dashed vertical line marks the cutoff between Semesters 1 & 2



The background of the slide features abstract, wavy, organic shapes in shades of blue and green. These shapes are layered, creating a sense of depth and movement. The word "Methodology" is centered in a dark blue, sans-serif font.

Methodology



Goal: Evaluate Mentoring Impact Fairly Across Subjects and Courses

Are there problems with using raw grades and attendance?

- Raw grades and attendance aren't directly comparable across subjects or teachers.
- Two students with the same raw score may be performing very differently within the context of their class.

What can we use instead?

- A standardized metric that puts all values on a common scale.
- It should account for differences in class difficulty, grading scale, and variation among peers.

The Answer: Z-Score

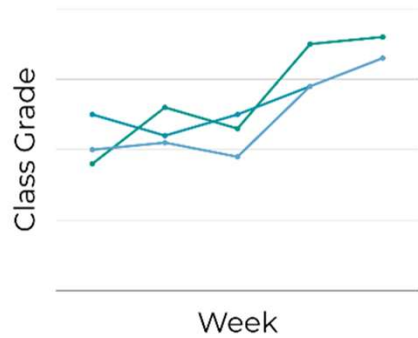
$$z_{itc} = \frac{X_{itc} - \mu_{tc}}{\sigma_{tc}}$$

- **Z_{itc}** : Z-score of grade or attendance of mentored student (i = individual student, t = week, c = class)
- **X_{itc}** : Grade or attendance of mentored student (i = individual student, t = week, c = class)
- **μ_{tc}** : Class grade or attendance average of non-mentored students
- **σ_{tc}** : Class standard deviation (how spread the grades or attendance is for that given week)

What does this student's Z-score tell us?

- We're tracking **how a mentored student's performance changes week by week**.
- A **Z-score** shows how that student compares to their classmates, using the **class average (μ)** and **standard deviation (σ)**.
- A **Z-score of 0** means the student is **right at the class average**.
- A **Z-score of +1** means they're **one standard deviation above average** (doing better than most peers that week), while **-1** means **one standard deviation below** (doing worse).

Interpreting Standardized GPA Over Time



Compare students across different classes/interventions and time windows

A whiteboard on a stand with the z-score formula written on it.

$$z = \frac{x - \mu}{\sigma}$$

Account for different course level difficulty and grade fluctuation due to nature of rolling grades



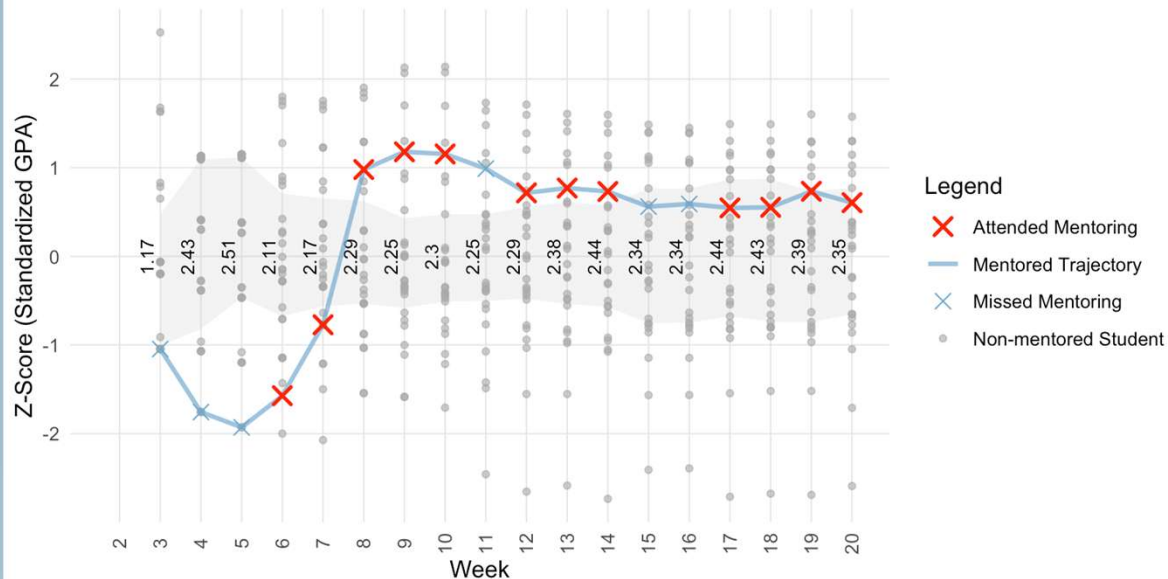
Observe relative performance changes rather than raw grades giving more context

Sample Graph Using Z-score Methodology

Student.ID	Course.Description	Course.Code	Class.Mean	Class.SD	Z.Score	Week	Attended
6	GEOMETRY A	MAT522 / 12	2.395	0.761	1.084	10	Yes
33	GEOMETRY A	MAT522 / 12	2.395	0.761	-1.032	10	Not Applicable
82	GEOMETRY A	MAT522 / 12	2.395	0.761	-0.611	10	Not Applicable
86	GEOMETRY A	MAT522 / 12	2.395	0.761	-0.020	10	Not Applicable
144	GEOMETRY A	MAT522 / 12	2.395	0.761	-0.558	10	Not Applicable
157	GEOMETRY A	MAT522 / 12	2.395	0.761	-0.545	10	Not Applicable
164	GEOMETRY A	MAT522 / 12	2.395	0.761	0.756	10	Not Applicable
166	GEOMETRY A	MAT522 / 12	2.395	0.761	-0.650	10	Not Applicable

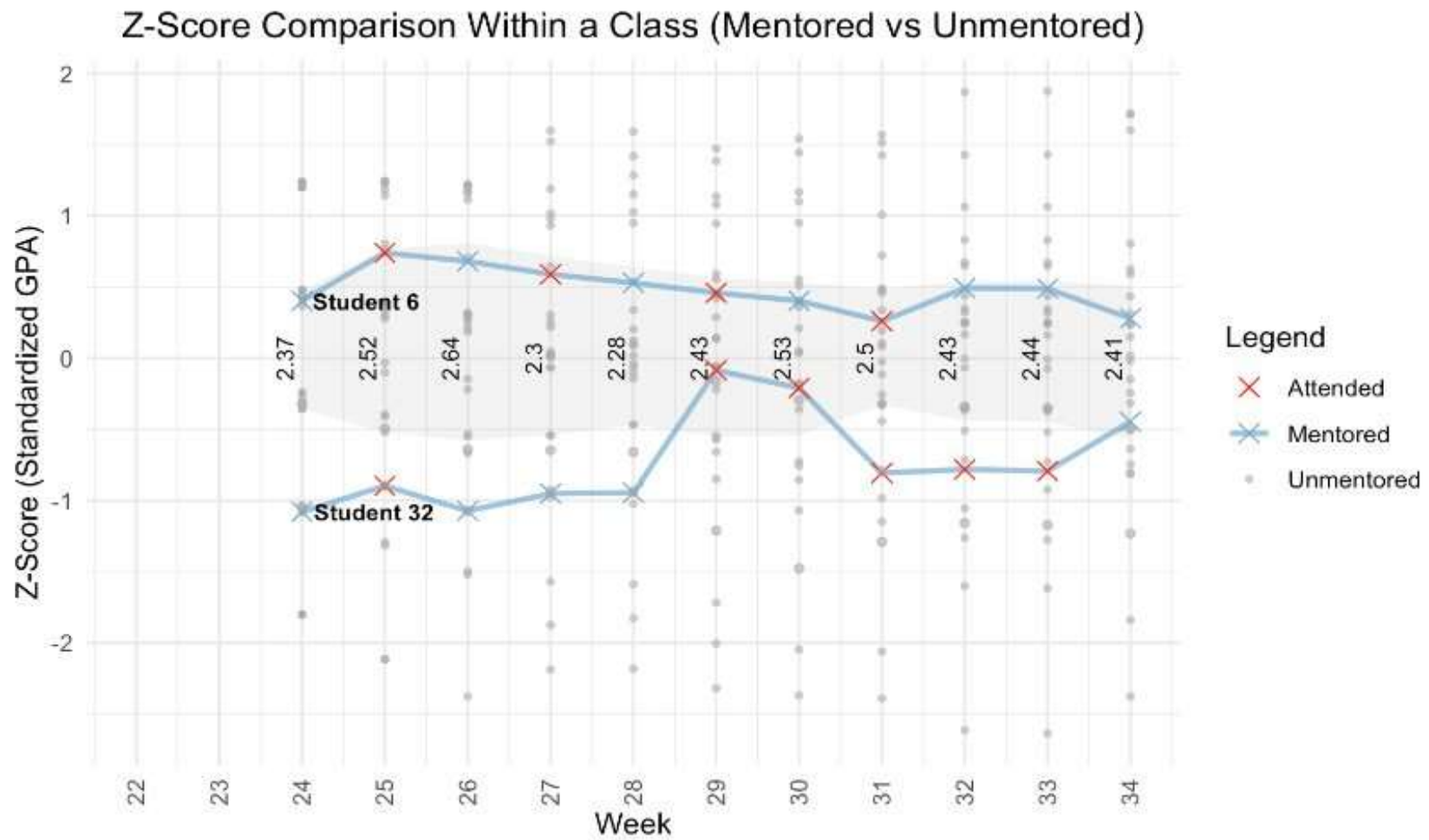
- This plot tracks the Z-score of Student 6—a mentored student—throughout the semester in a specific class.
- This class is Geometry A, taught by Rochelle Aiton during Period 2.
- Note: It has 25 total students.

Z-Score Trajectories by Week (Mentored vs. Non-Mentored)



- A red X indicates that Student 6 attended mentoring that week for that class.
- The blue line tracks the Student 6's weekly performance in this Geometry class.
- A blue X indicates that Student 6 did not attend mentoring that week for that class
- Gray dots represent non-mentored students' z-scores each week.
- The gray ribbon shows one standard deviation above and below the class average—roughly capturing where the middle 50% of students are performing.
- This view helps reveal class trends—like grade dips after difficult exams or signs of steady improvement.

Follow-Up Semester 2





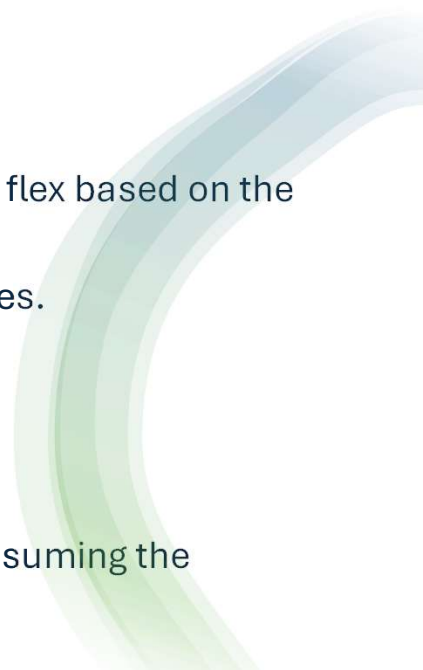
What is a GAM (Generalized Additive Model)?

A GAM helps us understand how different factors affect student outcomes — even when the patterns aren't straight lines.

How does it work?

- Instead of forcing a simple straight-line relationship, a GAM allows the model to curve and flex based on the data.
- It can capture more realistic trends, like gradual improvement, plateaus, or sudden changes.

Why do we use it?

- Mentoring might affect students differently over time.
 - A GAM lets us see how student grades (or attendance) change week by week — without assuming the change is always the same.
- 

MODEL FOR GRADES



```
Z.Score ~ Mentor.Course * Grade.Year + Mentor.Course * Course.Category  
s(Week.sem, k = 15) + Period + Semester + s(Running.Total..Class.,  
k = 6) + s(Total.Days.Absent, k = 6) + s(Student.ID, Week.sem,  
bs = "re") + s(Student.ID, bs = "re")
```

MODEL FOR ATTENDANCE



```
Absence.Z.Score ~ Mentor.Course * Grade.Year + Mentor.Course *  
Course.Category + s(Week.sem, k = 15) + Period + Semester +  
s(Running.Total..Class., k = 6) + s(Student.ID, Week.sem,  
bs = "re") + s(Student.ID, bs = "re")
```

Goal is to model and predict student performance by combining key factors

- Who the student is accounting for individual differences
- Whether they're mentored and in which subject/grade (interaction terms).
 - Interaction terms let the effect of mentoring vary depending on subject or grade.
 - Mentoring may affect students different in Math or English for example
- How far into the semester they are (smooth over Week.sem)
 - A smooth term lets model capture unique classroom patterns such as early struggles, mid semester dips, or late progress
- How many mentoring sessions for that class $s(\text{Running.Total..Class})$
- Period to see the significance of when classes are held
- Together, these terms let us see a layered story of how mentoring, attendance, and time in semester combine while recognizing the nuanced characteristics of a class

GAM Visualization Setup

Defining a “Typical” Student

To keep things simple and consistent, we used the following characteristics for a “typical” student in our analysis:

- 10th grade
- Period 1 class
- Semester 1
- Around 4 days absent (the average)

By keeping these the same, we were able to focus on how time and mentoring affect performance—without being influenced by differences in grade level or attendance.

Variables That Can Change in Our Model

These are the factors we allowed to vary in order to better understand the effects of mentoring:

- Week (1–20)
 - Modeled with $s(\text{Week}, k=15)$ to let the curve flexibly capture early-semester, midterm, and late-semester trends.
- Mentor.Course (Yes vs. No) × Course.Category
 - Interaction lets the “mentoring boost” differ by subject (Math vs. English vs. Science, etc.).
 - Math students might gain a larger jump from each session than Business students
- Running.Total..Class
 - Mentoring frequency changes by subject, so we adjust the gap accordingly

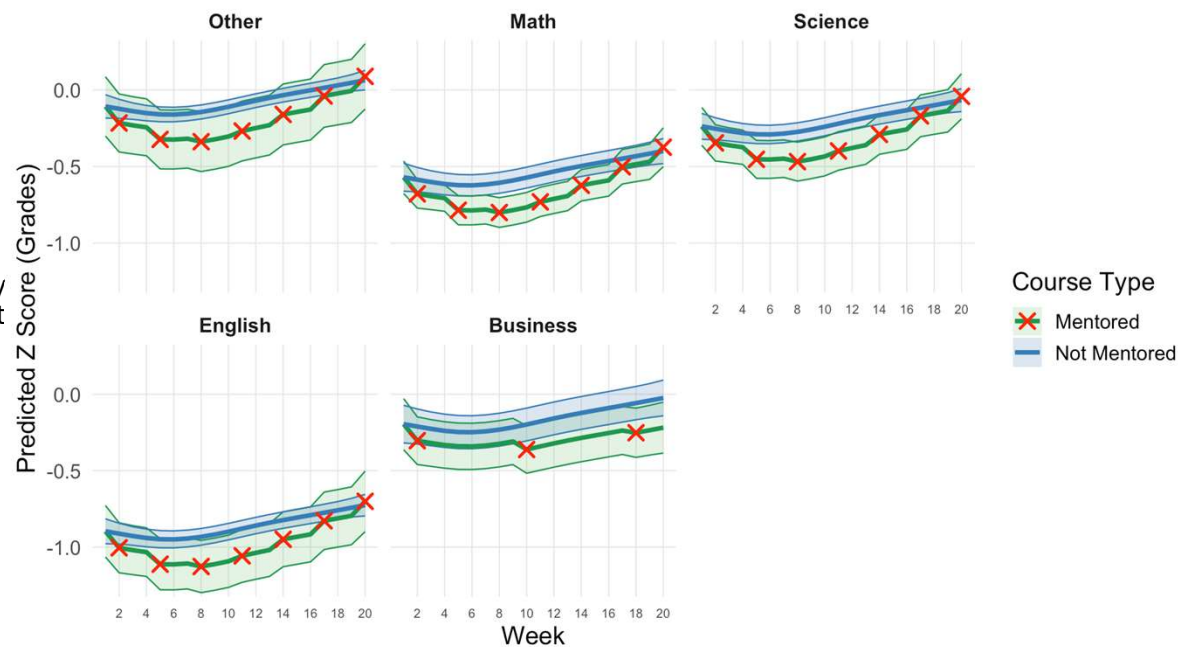


Results

Predicted Z Score Trajectories: Across All Subjects

- This graph shows "**what-if**" performance trajectories for a **mentoring program participant** under two scenarios:
 - **Green line:** receiving mentoring in the course
 - **Blue line:** not receiving mentoring in the course
- **Red X:** Represents evenly spaced mentoring sessions based on the **average number of sessions attended** by the average student. For example, if the average student attended 3 sessions, three evenly spaced red X's are plotted to illustrate typical mentoring frequency.
- Predictions are based on our **GAM model for grades**.
- Early mentoring sessions show modest improvement, with a cumulative effect leading to greater gains over time.

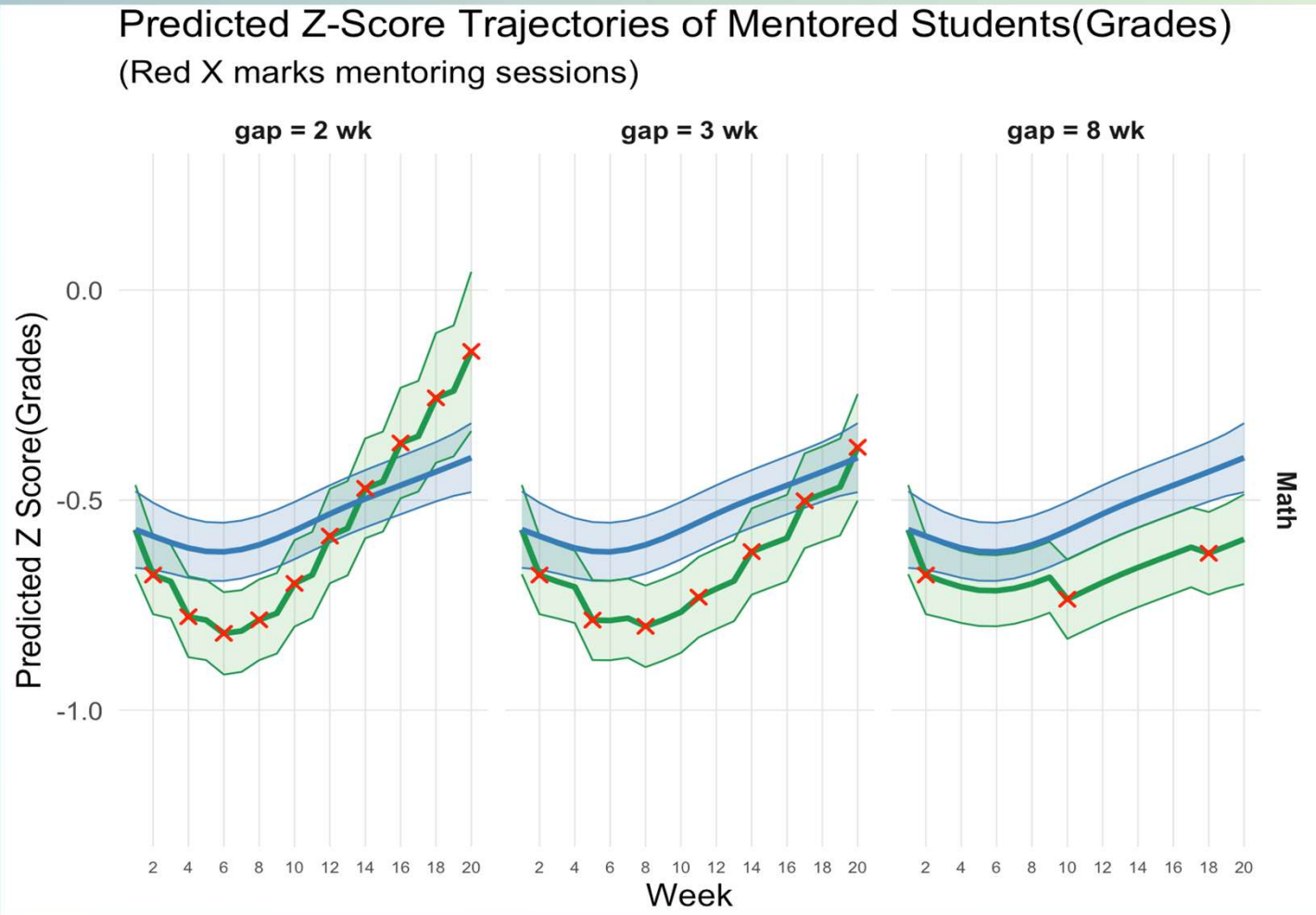
Predicted Z-Score Trajectories of Mentored Students (Grades)
(Red X marks mentoring sessions)



"Best", "Typical", and "Worst" Case Scenario: Math

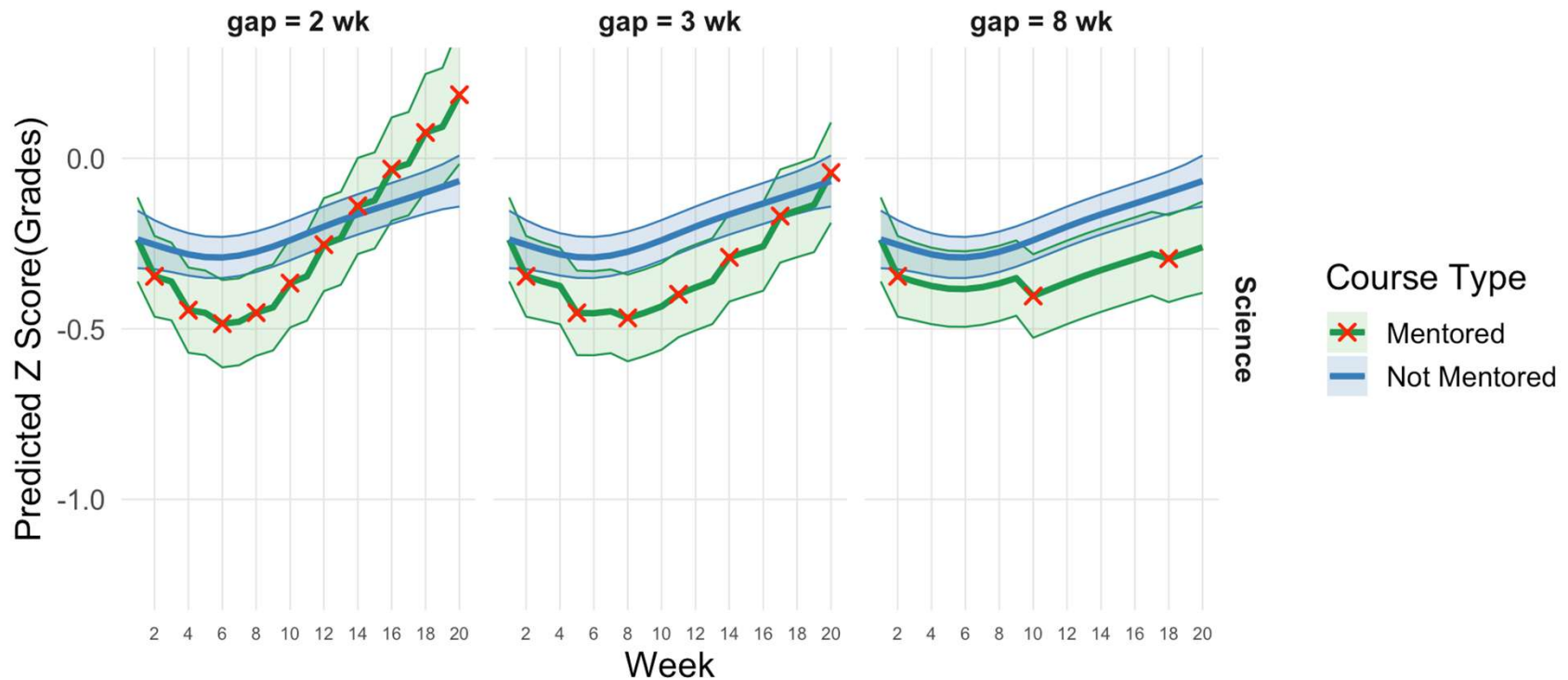
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- This graph shows a what if trajectory for a mentoring program participant when mentoring sessions are held every **2,3, or 8** weeks.
- Let's us predict performance based on how **often** a student attends a session



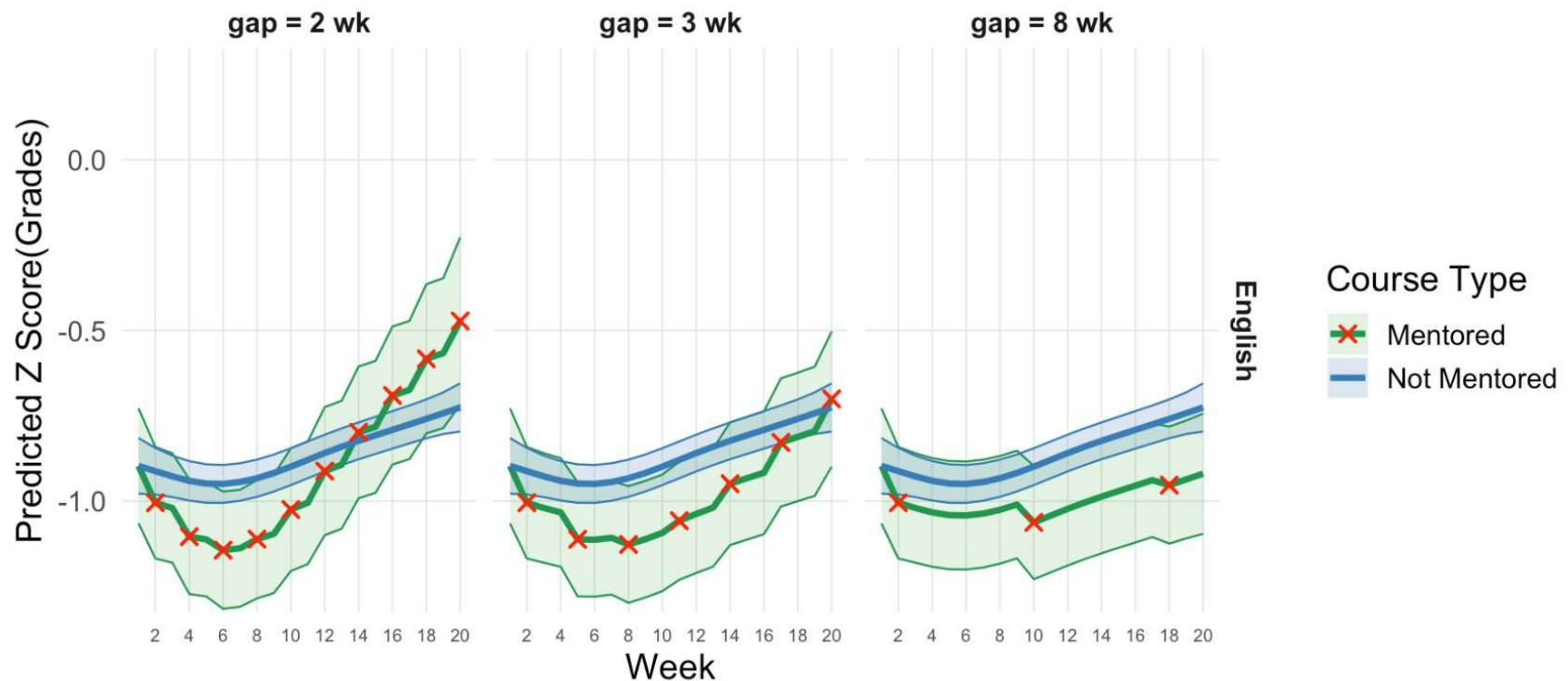
"Best", "Typical", and "Worst" Case Scenario: Science

Predicted Z-Score Trajectories of Mentored Students(Grades)
(Red X marks mentoring sessions)



"Best", "Typical", and "Worst" Case Scenario: English

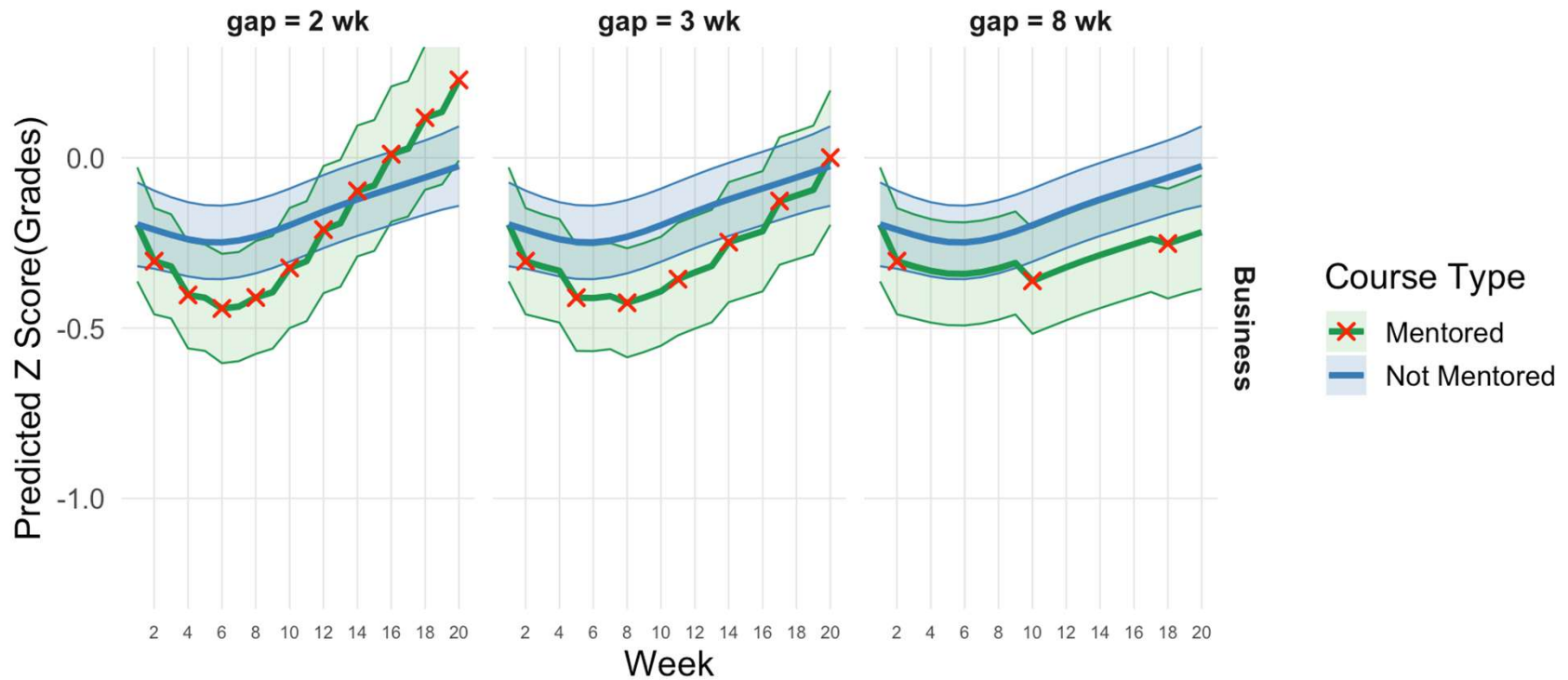
Predicted Z-Score Trajectories of Mentored Students(Grades)
(Red X marks mentoring sessions)



"Best", "Typical", and "Worst" Case Scenario: Business

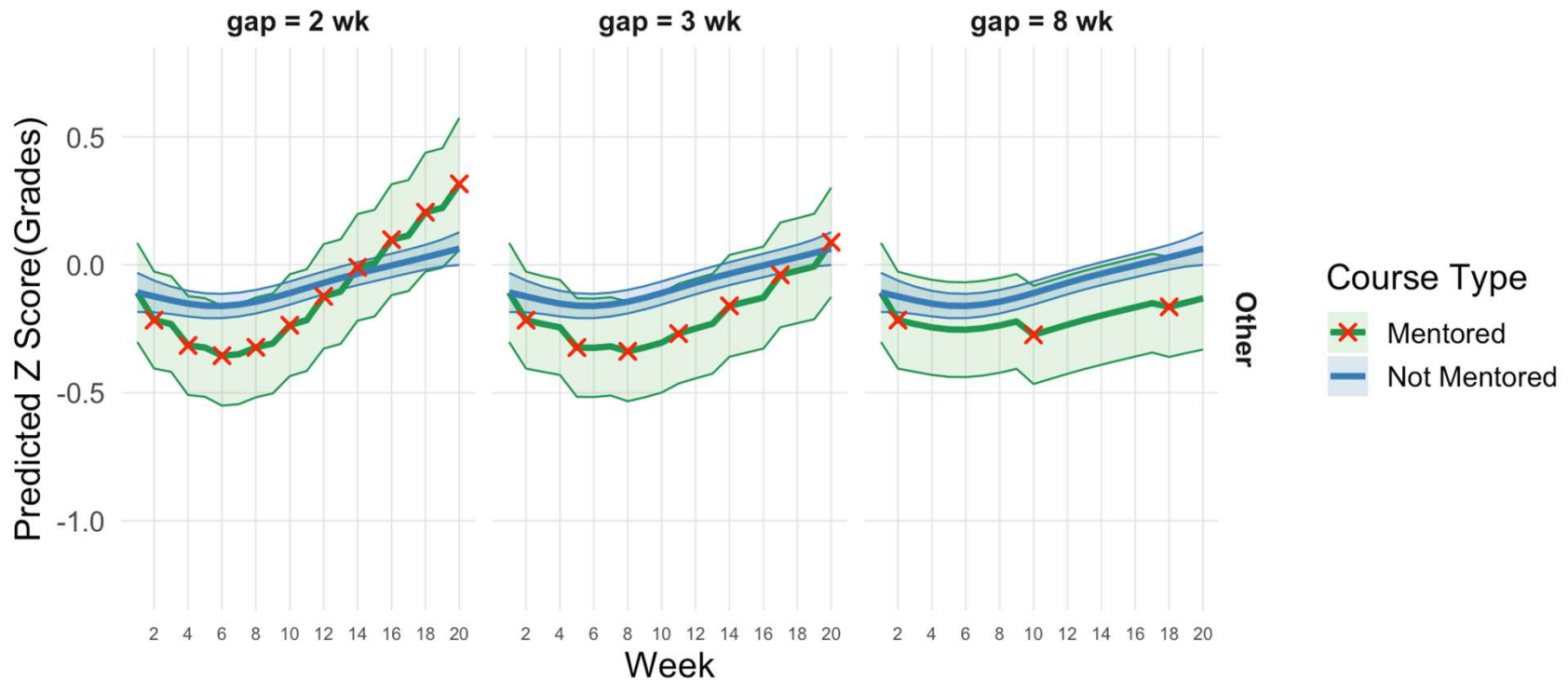
Predicted Z-Score Trajectories of Mentored Students(Grades)

(Red X marks mentoring sessions)



"Best", "Typical", and "Worst" Case Scenario: Other

Predicted Z-Score Trajectories of Mentored Students(Grades)
(Red X marks mentoring sessions)



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Conclusion

Conclusion

Mentoring Impact on Grades and Attendance

- Mentoring sessions have less of a short-term boost on z-score; instead, the more sessions accumulated, the more performance is improved over time
 - Consistency is key
- The flexibility of GAM captured the non-linear weekly trends
 - Notable dip in grades around weeks 8-10 that can be explained by midterms or assignments
 - Something you can do about this in the future: emphasize mentoring sessions around this week with group reviews or in-depth study guides
- Mentored students demonstrate greater overall improvement, suggesting that consistent support and mentoring contribute to sustained academic success.



Next Steps

Analyze data after week 34

Potential Issue: the presence of skewed data

- Grades and attendance is skewed when most students score very high or very low, instead of being evenly spread out.
- This can potentially affect the results, but it's unclear by how much.

Next Steps: what can be done to fix this

- Use percentiles instead of Z-scores (e.g., the student has a grade higher than 75% of other students in their class that week)



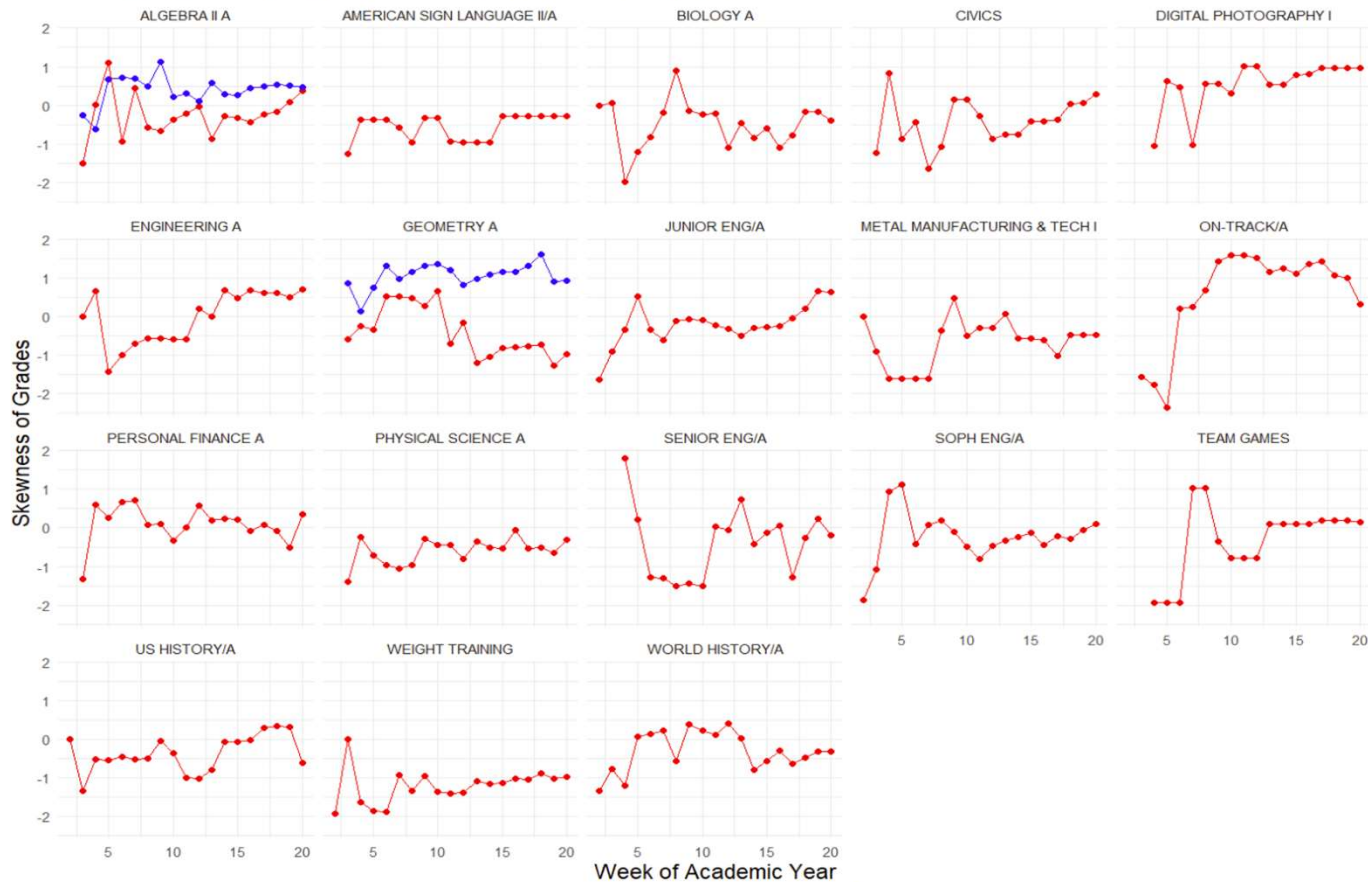
Thank you! Questions?

Appendix

What is the weekly skewness of the grades for mentored students across semester 1?

Weekly Skewness of Mentored Students Grades by Course Description (Semester 1)

Skewness calculated using $3 \times (\text{mean} - \text{median}) / \text{standard deviation}$



Mentored Course Indicator

- At least 1 student not mentored in this course (red)
- At least 1 student mentored in this course (blue)

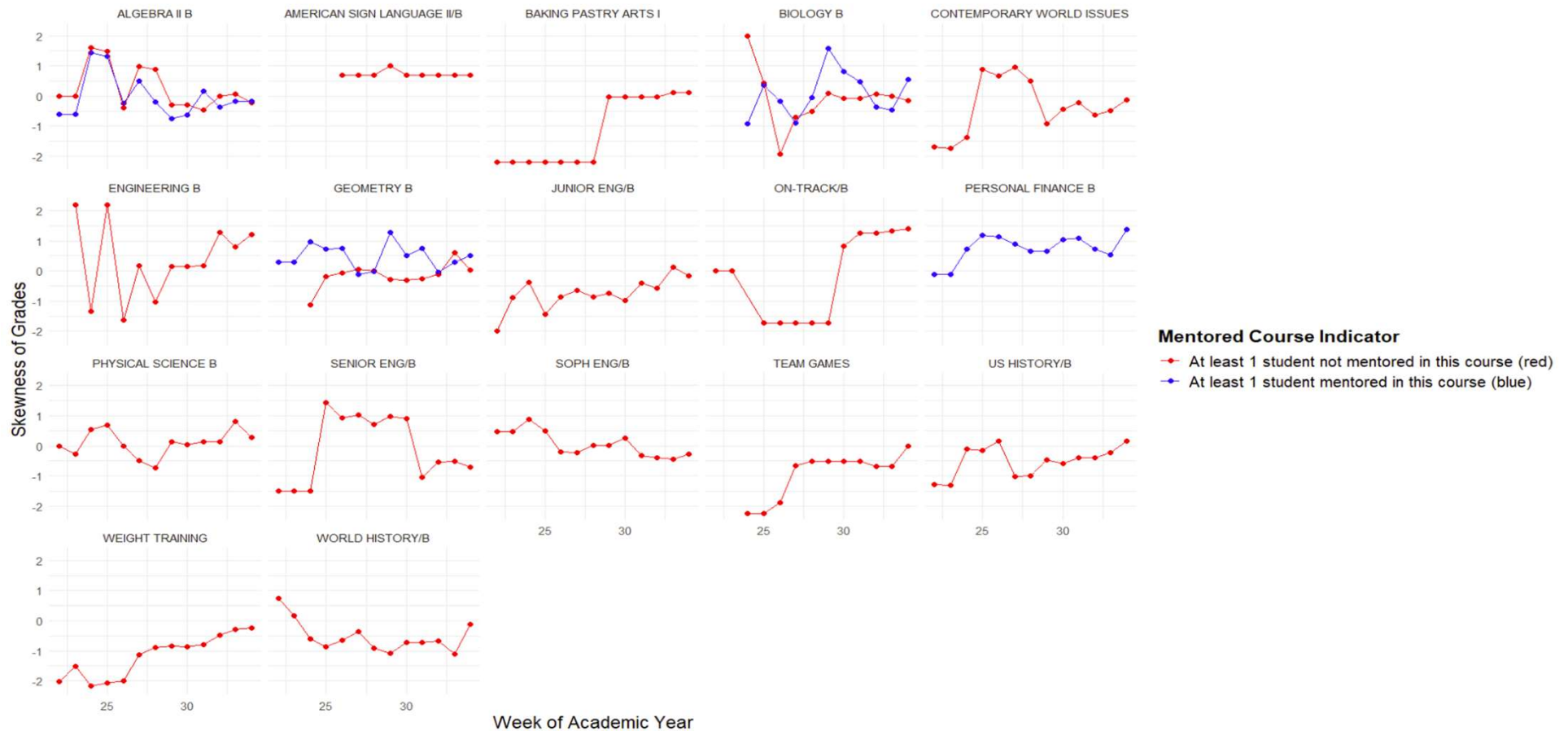
Week of Academic Year

Note: Courses with fewer than 5 mentored students were filtered out for reliable skewness calculation.

What is the weekly skewness of the grades for mentored students across semester 2?

Weekly Skewness of Mentored Students Grades by Course Description (Semester 2)

Skewness calculated using $3 \times (\text{mean} - \text{median}) / \text{standard deviation}$

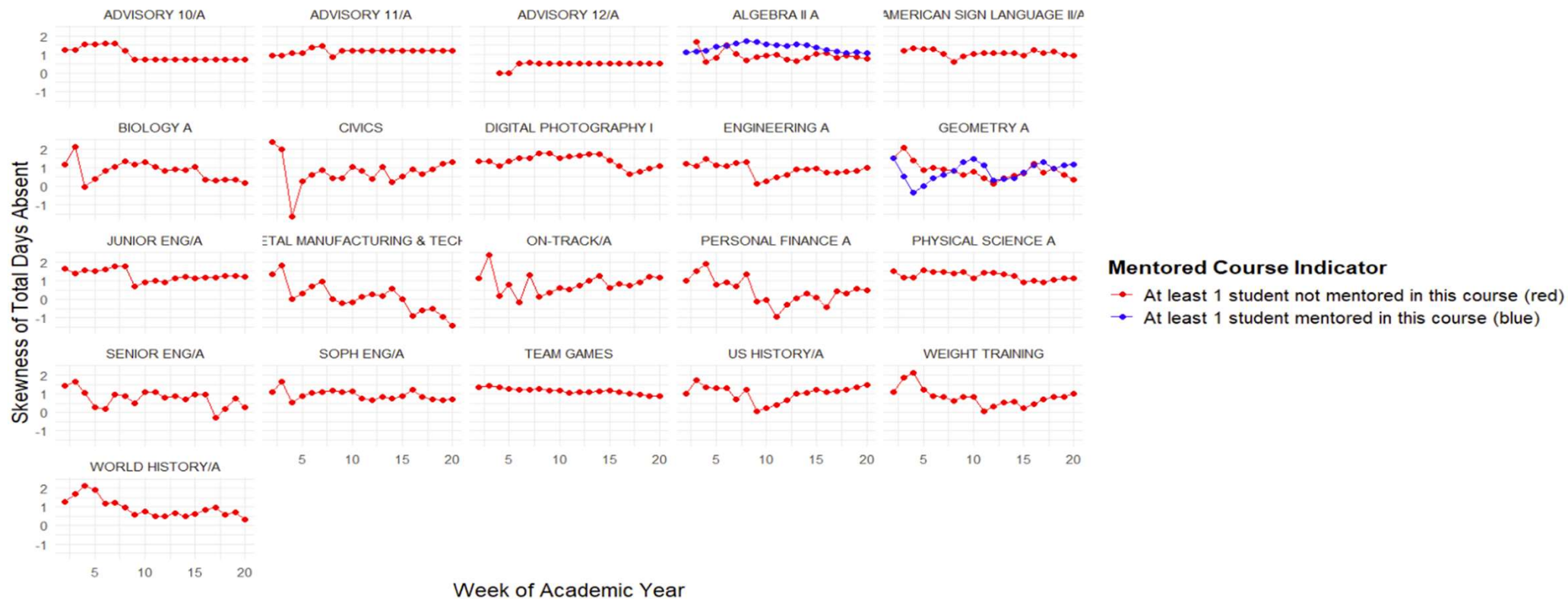


Note: Courses with fewer than 5 mentored students were filtered out for reliable skewness calculation.

What is the weekly skewness of the absences from class for mentored students across semester 1?

Weekly Skewness of Mentored Students' Absences by Course Description (Semester 1)

Skewness calculated using $3 \times (\text{mean} - \text{median}) / \text{standard deviation}$

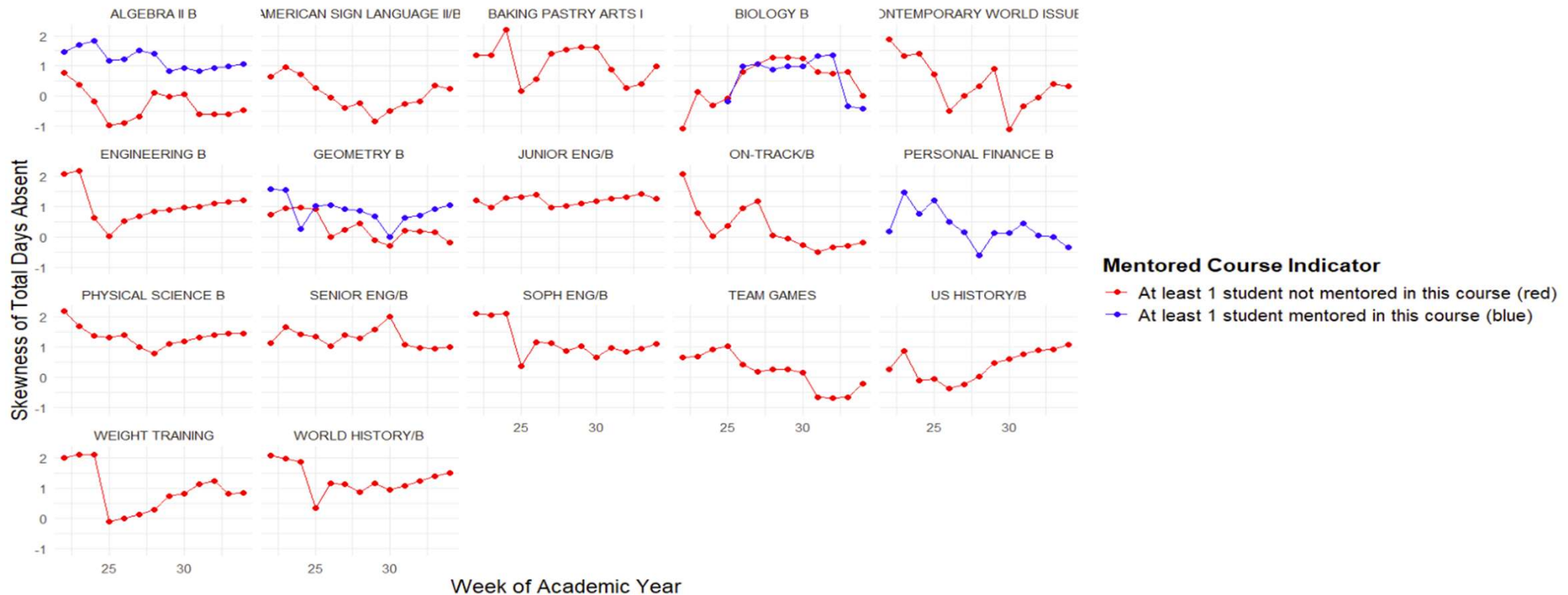


Note: Courses with fewer than 5 mentored students were filtered out for reliable skewness calculation.

What is the weekly skewness of the absences from class for mentored students across semester 2?

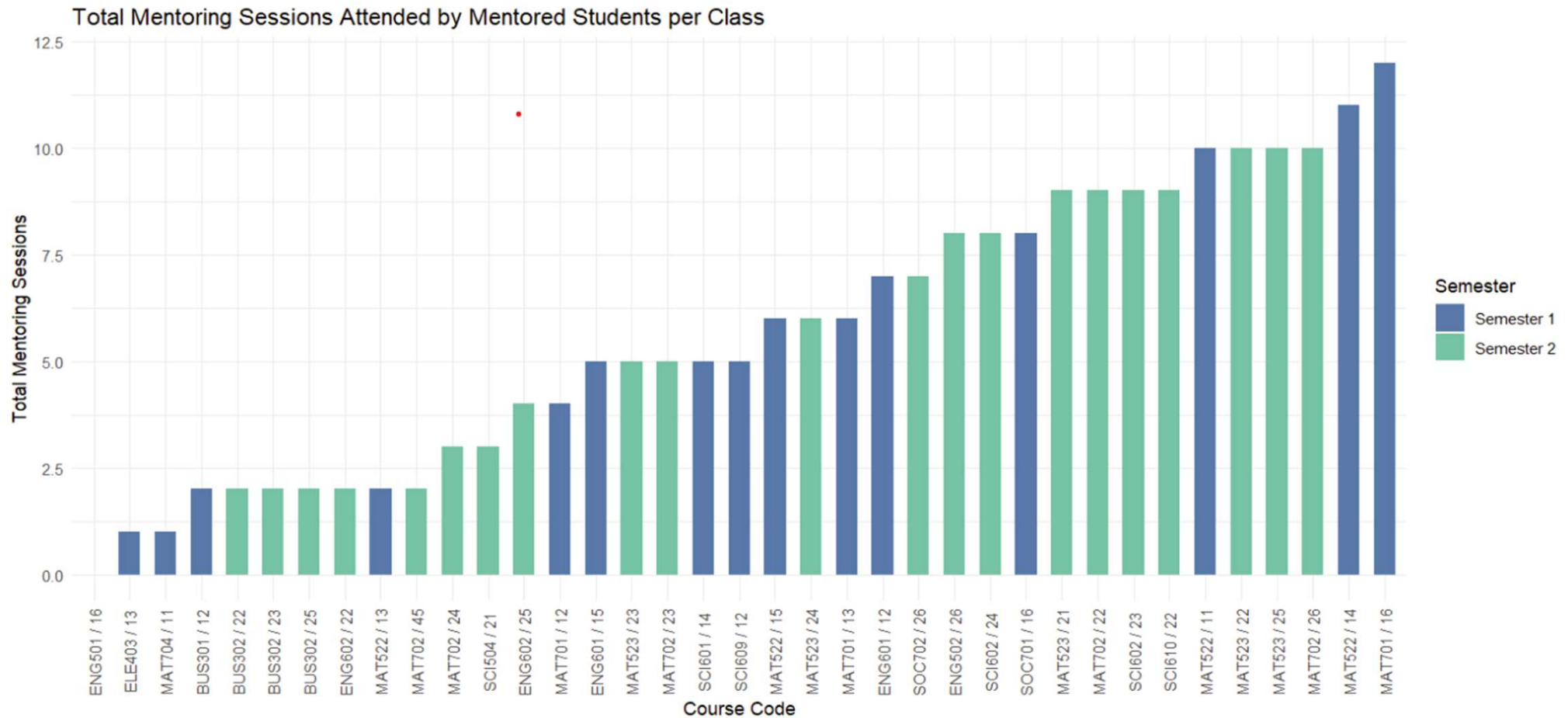
Weekly Skewness of Mentored Students' Absences by Course Description (Semester 2)

Skewness calculated using $3 \times (\text{mean} - \text{median}) / \text{standard deviation}$



Note: Courses with fewer than 5 mentored students were filtered out for reliable skewness calculation.

Which classes had the most mentoring sessions?



Modeling Results: Grades Model

(Intercept)	2.78e-06	***
Mentor.CourseYes	1.76e-09	***
Grade.Year	0.000966	***
Course.CategoryMath	0.017900	*
Course.CategoryScience	0.757475	
Course.CategoryEnglish	0.284328	
Course.CategoryBusiness	1.01e-07	***
Week.sem	9.00e-16	***
Period	0.012746	*
SemesterS2	0.007846	**
Mentor.CourseYes:Grade.Year	6.41e-13	***
Mentor.CourseYes:Course.CategoryMath	0.228717	
Mentor.CourseYes:Course.CategoryScience	0.845414	
Mentor.CourseYes:Course.CategoryEnglish	2.26e-12	***
Mentor.CourseYes:Course.CategoryBusiness	0.003747	**
Course.CategoryMath:Week.sem	2.50e-06	***
Course.CategoryScience:Week.sem	0.000243	***
Course.CategoryEnglish:Week.sem	0.282610	
Course.CategoryBusiness:Week.sem	0.203428	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1		
Approximate significance of smooth terms:		
	edf	Ref.df F p-value
s(Running.Total..Class.)	3.3431980	3.904 8.821 6.06e-07 ***
s(Total.Days.Absent)	4.2282165	4.677 182.041 < 2e-16 ***
s(Student.ID,Week.sem)	0.0005679	1.000 0.000 0.743
s(Student.ID)	0.3469445	1.000 0.532 0.215

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1		
R-sq.(adj) = 0.177 Deviance explained = 17.9%		
-REML = 10915 Scale est. = 0.48474 n = 10264		

Modeling Results: Attendance Model

	Pr(> t)	
(Intercept)	2.04e-12 ***	
Mentor.CourseYes	0.679842	
Grade.Year	6.72e-14 ***	
Course.CategoryMath	0.005540 **	
Course.CategoryScience	0.212640	
Course.CategoryEnglish	2.71e-05 ***	
Course.CategoryBusiness	4.28e-10 ***	
Period	0.000282 ***	
SemesterS2	0.000795 ***	
Mentor.CourseYes:Grade.Year	0.859091	
Mentor.CourseYes:Course.CategoryMath	0.007596 **	
Mentor.CourseYes:Course.CategoryScience	1.20e-06 ***	
Mentor.CourseYes:Course.CategoryEnglish	0.005627 **	
Mentor.CourseYes:Course.CategoryBusiness	0.040217 *	

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Week.sem):Course.CategoryOther	1.0057	1.011	3.037	0.0814 .
s(Week.sem):Course.CategoryMath	1.1924	1.360	3.606	0.0326 *
s(Week.sem):Course.CategoryScience	1.0050	1.010	3.070	0.0797 .
s(Week.sem):Course.CategoryEnglish	1.0041	1.008	1.654	0.1982
s(Week.sem):Course.CategoryBusiness	1.1045	1.201	0.072	0.9298
s(Running.Total..Class.)	1.0013	1.003	5.554	0.0185 *
s(Student.ID,Week.sem)	0.9622	1.000	175.898	2.57e-07 ***
s(Student.ID)	0.5288	1.000	8.450	0.1452

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.0301 Deviance explained = 3.21%
-REML = 12041 Scale est. = 0.59728 n = 10327
